

INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY SHIBPUR

Department of Mechanical Engineering



Optimization of Heat Flux to Reduce Crack Breakout in TSCR at Tata Steel

*An Internship Report submitted in partial fulfilment of the requirements for the Vocational Training /
Internship Programme*

Submitted by

Sneha Mishra

Under the Guidance of

Mr. Alok Kumar

Head Caster, LD3 TSCR Department At Tata Steel

Department of Mechanical Engineering

Indian Institute of Engineering Science and Technology (IIEST), Shibpur

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Sneha Mishra

Department of Mechanical Engineering

Indian Institute of Engineering Science and Technology (IEST), Shibpur

ABSTRACT

Continuous casting is a critical process in modern steel manufacturing, where efficient heat transfer during solidification plays a vital role in determining the quality of the final product. In the Thin Slab Casting and Rolling (TSCR) process, improper heat extraction in the mould can lead to the formation of longitudinal surface cracks and, in severe cases, breakout, resulting in production losses, reduced product quality, and increased operational costs. Therefore, understanding the factors influencing heat flux and optimizing the casting process are essential for improving process stability and ensuring the production of defect-free steel slabs.

This internship project, carried out at Tata Steel Limited, is focused on the optimization of heat flux to reduce crack breakout in the TSCR process. The study commenced with an extensive literature review to understand the continuous casting process, mechanisms of heat transfer, shell formation, longitudinal crack formation, breakout phenomena, mould flux behaviour, and the influence of various process parameters on heat flux. Plant visits were conducted to gain practical exposure to the TSCR process and to understand the operation of the mould, secondary cooling system, casting powder, and breakout detection system.

Historical plant process data were collected and analyzed to investigate the relationship between average heat flux and various process parameters. The dataset was preprocessed by handling missing values, calculating average heat flux, and selecting the most relevant variables for analysis. Exploratory Data Analysis (EDA), scatter plot analysis, correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis were performed to identify the key process parameters affecting heat flux.

As a part of the project, trial mould and casting powder experiments were also initiated to improve the heat transfer characteristics of the mould. The experimental data obtained from these trials will be compared with the existing plant operating conditions to evaluate their effectiveness in improving heat flux distribution and reducing the likelihood of crack breakout. Furthermore, breakout records and Design of Experiments (DOE) will be utilized to validate the findings and identify optimal operating conditions for improved casting performance.

The outcomes of this study provide a comprehensive understanding of the process parameters affecting heat flux in the TSCR process and demonstrate the application of statistical analysis and machine learning techniques in industrial process optimization. The findings are expected to contribute towards enhancing process stability, minimizing crack breakout, and improving product quality and operational efficiency in continuous casting operations at Tata Steel Limited.

CHAPTER 1

INTRODUCTION



1.1 About Tata Steel

Tata Steel Limited is one of the world's leading steel manufacturing companies and a flagship enterprise of the Tata Group. Established in **1907** by **Jamsetji Nusserwanji Tata**, the company has played a significant role in the growth of India's industrial sector and is recognized for its commitment to quality, innovation, sustainability, and technological excellence. Headquartered in Mumbai, India, Tata Steel operates integrated steel manufacturing facilities in India as well as in several international locations.

The Jamshedpur Works, commissioned in 1912, is India's first integrated steel plant and continues to be one of the most technologically advanced steel manufacturing facilities in the country. The plant consists of various production units, including Raw Material Handling, Coke Ovens, Sinter Plant, Blast Furnace, Steel Melting Shop (SMS), Thin Slab Casting and Rolling (TSCR), Hot Strip Mill, Cold Rolling Mill, Galvanizing Lines, and By-Product Management facilities. These units work together to ensure efficient steel production while maintaining high standards of product quality and operational safety.

Tata Steel has consistently focused on digital transformation and the adoption of Industry 4.0 technologies to improve manufacturing efficiency and process reliability.

Advanced process automation, real-time monitoring systems, predictive analytics, and data-driven decision-making are extensively employed across its production facilities. These technologies help optimize process parameters, improve product quality, reduce operational costs, and enhance plant productivity.

The company is equally committed to sustainable steel manufacturing. Through efficient resource utilization, waste recycling, energy conservation, and by-product recovery, Tata Steel strives to minimize its environmental impact. Various by-products generated during steelmaking are processed into useful commercial products, supporting the company's vision of a circular economy and sustainable industrial development.



Fig 1. Panoramic View of Tata Steel Limited, Jamshedpur

This internship was carried out in the **Thin Slab Casting and Rolling (TSCR) Department** of Tata Steel, Jamshedpur. The project focuses on the optimization of heat flux to reduce crack breakout during continuous casting. The study combines literature review, plant observations, statistical analysis, and machine learning techniques to identify the critical process parameters affecting heat transfer within the mould and to support process optimization for improved casting performance and product quality.

The opportunity to undertake this internship at Tata Steel provided valuable exposure to industrial steel manufacturing practices and the application of data-driven techniques for process optimization. Through interactions with plant engineers, observation of real-time casting operations, and analysis of historical process data, a deeper understanding was developed of the challenges involved in maintaining stable heat transfer during continuous casting. The knowledge and experience gained during this internship form the foundation for the analyses and discussions presented in the subsequent chapters of this report.

1.2 Continuous Casting

Continuous casting is one of the most significant advancements in the steel manufacturing industry. It is a continuous production process in which molten steel is directly solidified into semi-finished products such as slabs, blooms, or billets. Compared with the conventional ingot casting process, continuous casting offers higher productivity, improved product quality, better yield, lower energy consumption, and reduced production costs. Today, almost all modern integrated steel plants employ continuous casting due to its efficiency and reliability.

The process begins after molten steel is produced in the Steel Melting Shop (SMS). The molten steel is first poured into a ladle, which serves as a transport vessel carrying the steel to the continuous casting machine. The ladle then transfers the molten steel into a tundish, which acts as an intermediate reservoir. The tundish ensures a continuous and controlled flow of molten steel to the mould while also allowing non-metallic inclusions to float to the surface, thereby improving steel cleanliness.

From the tundish, molten steel enters the water-cooled copper mould through a Submerged Entry Nozzle (SEN). The mould is the most critical component of the continuous casting machine because it is the primary heat extraction zone. As molten steel comes into contact with the cooled copper walls of the mould, rapid heat transfer occurs, resulting in the formation of a thin solid shell around the liquid core. This shell must possess sufficient strength to withstand the ferrostatic pressure of the molten steel contained inside it.

After leaving the mould, the partially solidified strand enters the secondary cooling zone, where water sprays and support rollers provide controlled cooling and mechanical support. During this stage, the shell thickness gradually increases while the liquid core continues to solidify. The rollers prevent deformation of the strand and ensure dimensional stability throughout the casting process.

Once complete solidification is achieved, the strand passes through the straightening section, where it is gradually straightened from its curved path. The continuously cast strand is then cut into slabs of predetermined lengths using automatic cutting equipment before being transported to the rolling mill for further processing.

The quality of continuously cast steel depends largely on maintaining stable heat transfer during solidification. Excessive heat extraction may generate high thermal stresses, while insufficient cooling may produce an inadequate shell thickness. Both conditions can lead to surface defects, internal cracks, or, in severe cases, breakout, where molten steel escapes through the solid shell, causing production loss and equipment damage. Therefore, maintaining an optimum heat transfer rate is essential for ensuring safe operation and producing defect-free slabs.

Several process variables influence heat transfer during continuous casting, including casting speed, mould level, tundish temperature, superheat, mould cooling water flow rate, mould cooling water temperature, casting powder characteristics, oscillation parameters, and steel

grade. Continuous monitoring and optimization of these parameters are necessary to achieve uniform solidification, improve product quality, and minimize casting defects.

With the advancement of digital technologies, modern steel plants continuously record thousands of process parameters during casting. These large datasets can be analyzed using statistical methods and machine learning techniques to identify the variables that most significantly affect heat transfer and casting quality. Such data-driven approaches enable engineers to optimize process conditions, improve operational stability, and reduce the occurrence of defects such as longitudinal cracks and breakout. The present study follows a similar methodology to investigate the influence of key process parameters on heat flux in the TSCR process at Tata Steel

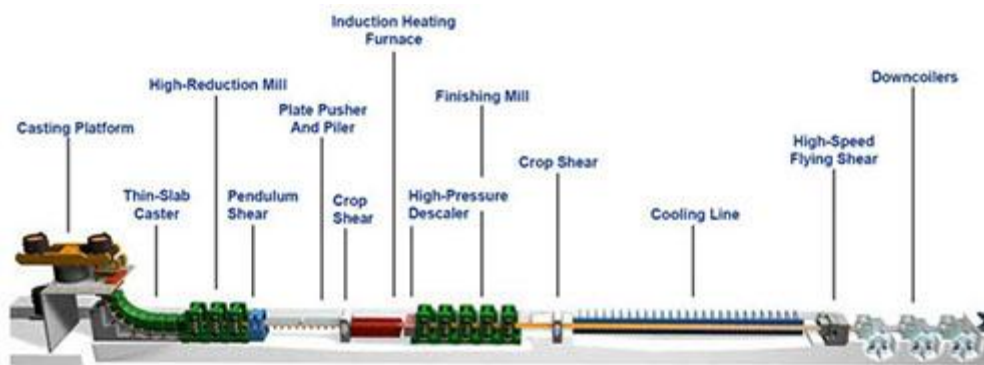


Fig 2. Schematic representation of the thin slab caster..

1.3 Thin Slab Casting and Rolling (TSCR)

Thin Slab Casting and Rolling (TSCR) is an advanced steel manufacturing technology that integrates continuous casting and hot rolling into a single continuous production line. Unlike the conventional steel production route, where cast slabs are cooled, stored, and reheated before rolling, the TSCR process enables the direct transfer of thin slabs from the caster to the rolling mill while they retain a significant amount of thermal energy. This integration not only improves production efficiency but also reduces energy consumption, operational costs, and environmental emissions.

The TSCR process begins with the production of molten steel in the Steel Melting Shop (SMS). The molten steel is transported in a ladle to the casting machine and subsequently transferred to the tundish, which acts as an intermediate reservoir. The tundish ensures a stable and continuous flow of molten steel into the mould while minimizing turbulence and allowing non-metallic inclusions to float to the slag layer. From the tundish, molten steel enters the water-cooled copper mould through the Submerged Entry Nozzle (SEN), where rapid heat extraction initiates solidification and forms a thin solid shell around the liquid core.

As the partially solidified strand exits the mould, it passes through a series of support rollers and the secondary cooling zone, where precisely controlled water sprays continue the solidification process. The shell gradually thickens until complete solidification is achieved while the rollers maintain the strand geometry and prevent deformation. The fully solidified thin slab is then transferred directly to the hot rolling mill without undergoing a separate reheating stage. The residual heat contained within the slab is effectively utilized during rolling, resulting in significant fuel savings and improved process efficiency.

One of the major advantages of TSCR technology is its ability to produce high-quality hot rolled steel with a shorter production cycle and lower manufacturing costs. The elimination of intermediate slab storage and reheating substantially reduces energy requirements and carbon emissions while improving metal yield and overall plant productivity. These advantages have made TSCR one of the most preferred casting technologies in modern integrated steel plants.

Despite its numerous advantages, the TSCR process demands strict control of operating conditions throughout casting and rolling. Parameters such as casting speed, mould level, tundish temperature, superheat, mould cooling water flow rate, mould cooling water temperature, casting powder properties, and heat flux directly influence shell growth, solidification behaviour, and surface quality of the slab. Even minor deviations in these parameters can disturb heat transfer within the mould, resulting in non-uniform shell formation, longitudinal surface cracks, or, in severe cases, breakout. Therefore, continuous monitoring and optimization of these process variables are essential to ensure safe operation and consistent product quality.

The present internship was carried out in the **TSCR Department of Tata Steel, Jamshedpur**, with the objective of understanding the factors influencing mould heat flux and their role in crack breakout. The study involved detailed literature review, plant visits, process observation, analysis of historical plant data, and the application of statistical and machine learning techniques to identify the key process parameters governing heat transfer during continuous casting. The findings of this work aim to support process optimization and contribute towards improving casting stability, reducing breakout incidents, and enhancing the overall productivity of the TSCR process.

1.3.1 Major Components of the TSCR Process

The Thin Slab Casting and Rolling process consists of several interconnected units that work in synchronization to ensure continuous production of high-quality steel. Each component performs a specific function and contributes to maintaining stable casting conditions.

Ladle

The ladle receives molten steel from the Steel Melting Shop (SMS) and transports it to the continuous casting machine. Besides transportation, the ladle acts as a temporary storage vessel that helps maintain the temperature of molten steel before casting. Ladles are lined with refractory materials to withstand extremely high temperatures, typically exceeding **1600°C**.



Fig 3. Pouring of Molten Steel from Ladle during Continuous Casting

Tundish

The tundish serves as an intermediate vessel between the ladle and the mould. Its primary function is to provide a continuous and stable flow of molten steel into the mould during casting. The tundish also promotes the separation of non-metallic inclusions, thereby improving steel cleanliness. Flow control devices such as dams, weirs, and turbulence inhibitors are often used inside the tundish to improve flow characteristics and minimize turbulence.

Submerged Entry Nozzle (SEN)

The Submerged Entry Nozzle transfers molten steel from the tundish into the mould below the steel surface. By introducing steel beneath the slag layer, the SEN prevents re-oxidation of molten steel and reduces turbulence inside the mould. The design and immersion depth of the SEN significantly influence flow patterns, inclusion movement, and heat transfer.

Copper Mould

The water-cooled copper mould is the most critical component of the TSCR process. It extracts heat rapidly from the molten steel, initiating solidification and forming the initial solid shell.

Since the shell produced inside the mould supports the liquid steel during casting, maintaining stable heat transfer within the mould is essential. Variations in heat extraction may result in uneven shell thickness, longitudinal surface cracks, or breakout.

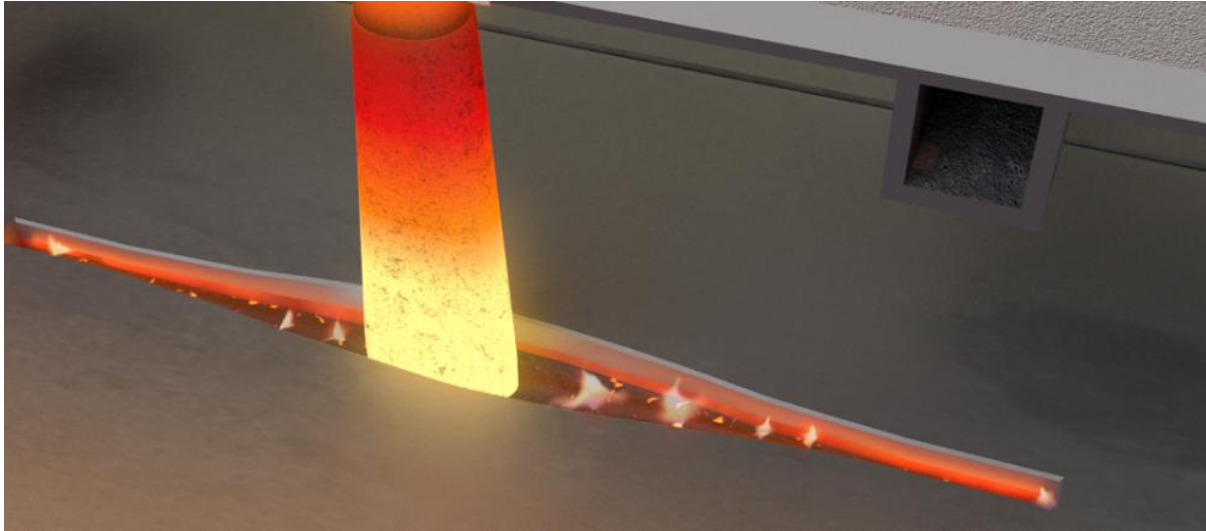


Fig 4.Copper Mould

Secondary Cooling Zone

After leaving the mould, the strand enters the secondary cooling zone, where water sprays remove the remaining heat from the steel. The cooling intensity is carefully controlled to avoid excessive thermal stresses while ensuring complete solidification. Rollers positioned along the strand provide mechanical support and prevent bulging of the partially solidified shell.

Straightening and Cutting

Once the strand is fully solidified, it passes through the straightening unit, where it is gradually converted from the curved casting path into a straight slab.

1.3.2 Advantages of TSCR Technology

The adoption of Thin Slab Casting and Rolling technology offers several technical, economic, and environmental advantages over the conventional slab casting process.

- Elimination of slab reheating significantly reduces fuel consumption.
- Higher thermal efficiency due to direct hot charging of slabs.
- Reduction in production time through continuous processing.
- Improved material yield owing to lower handling losses.
- Lower greenhouse gas emissions because of reduced fuel requirements.
- Enhanced product quality through better process control.
- Reduced operational and maintenance costs.

- Higher plant productivity due to integrated casting and rolling operations.
- Better dimensional accuracy and improved surface finish of rolled products.
- Increased flexibility in producing different steel grades.

These advantages have made TSCR one of the most preferred technologies for manufacturing flat steel products in modern integrated steel plants.

1.3.3 Challenges in the TSCR Process

Despite its numerous benefits, the TSCR process presents several operational challenges that require continuous monitoring and optimization.

The rapid solidification occurring within the mould makes heat transfer highly sensitive to changes in operating conditions. Small variations in casting speed, mould level, mould oscillation, casting powder behaviour, or cooling water conditions may significantly alter the heat flux. Such variations influence shell growth and can increase the likelihood of surface defects and breakout.

Maintaining a uniform heat flux throughout the mould is particularly important because excessive heat extraction can produce thermal stresses and crack formation, whereas insufficient heat extraction may lead to inadequate shell thickness. Therefore, balancing the heat removal rate is essential for ensuring safe operation and high product quality.

Furthermore, the properties of mould flux play a significant role in lubrication and heat transfer. Variations in flux melting characteristics, viscosity, crystallization behaviour, and consumption rate can alter the heat transfer between the mould and the solidifying shell. Consequently, optimization of mould powder is an active area of research in modern steel plants.

1.3.4 Relevance of the Present Study

The present internship project focuses on understanding the relationship between operating parameters and mould heat flux in the TSCR process at Tata Steel. Since heat flux directly governs shell growth during the initial stages of solidification, identifying the factors affecting heat transfer is essential for reducing longitudinal cracks and preventing breakout.

During the internship, plant process data were collected and analyzed using statistical techniques and machine learning methods. Scatter plot analysis, correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis were carried out to identify the process parameters having the greatest influence on average heat flux.

Parameters such as **casting speed, superheat, and mould level** were found to play a significant role in heat transfer behaviour.

In addition to historical data analysis, plant visits were conducted to understand the operational aspects of the TSCR process. Observations from these visits provided practical insight into

mould operation, secondary cooling, breakout detection systems, and the influence of casting powder on heat transfer.

Furthermore, trial mould and casting powder experiments were initiated to evaluate improvements in heat transfer characteristics. The analysis of these trial results, along with Design of Experiments (DOE), forms the final stage of this study and aims to recommend process conditions that minimize crack breakout while maintaining stable casting performance.

1.4 Heat Transfer and Heat Flux in Continuous Casting

Heat transfer is one of the most critical phenomena governing the continuous casting process. The quality of the cast slab, shell growth, solidification behaviour, and occurrence of casting defects are all strongly influenced by the rate at which heat is extracted from the molten steel. During continuous casting, molten steel enters the water-cooled copper mould at a temperature exceeding **1539°C**. As soon as it comes into contact with the mould walls, heat is rapidly transferred from the molten steel to the copper mould, initiating the solidification process.

The heat extracted in the mould is subsequently carried away by the circulating cooling water flowing through the mould plates. This continuous removal of heat leads to the formation of a solid shell around the liquid steel core. The thickness and strength of this shell are of utmost importance because it must withstand the ferrostatic pressure exerted by the molten steel during its passage through the casting machine. If the shell is too thin or develops weak regions, the liquid steel may rupture the shell, resulting in a breakout.

Heat transfer during continuous casting occurs through three primary mechanisms: conduction, convection, and radiation. Conduction is the dominant mode of heat transfer from the solidifying shell to the copper mould walls. Convective heat transfer occurs within the molten steel and in the cooling water circulating through the mould channels, while radiative heat transfer contributes to heat loss from the strand surface after it exits the mould. The combined effect of these mechanisms determines the overall cooling rate and solidification behaviour of the cast strand.

Heat flux is defined as the amount of heat transferred per unit surface area per unit time and is generally expressed in **MW/m²** or **kW/m²**. It serves as an important indicator of the efficiency of heat extraction from the molten steel. In industrial continuous casting operations, heat flux is continuously monitored because it reflects the thermal interaction between the mould and the solidifying shell. A stable heat flux indicates uniform shell growth and stable casting conditions, whereas sudden fluctuations may indicate changes in lubrication, mould powder behaviour, shell sticking, or abnormal casting conditions.

The magnitude of heat flux is influenced by several process parameters, including casting speed, superheat, mould level, mould cooling water flow rate, mould cooling water temperature, mould oscillation, mould powder properties, steel grade, and shell thickness. These variables interact with one another, making heat transfer a complex process. Consequently, statistical techniques and machine learning methods are increasingly employed to identify the dominant parameters influencing heat flux.

Maintaining an optimum heat flux is essential for producing high-quality steel slabs. Excessively high heat flux may generate severe thermal stresses, increasing the likelihood of longitudinal surface cracks. On the other hand, low heat flux may result in insufficient shell growth, making the strand susceptible to bulging or breakout. Therefore, achieving a balanced and stable heat transfer rate is one of the primary objectives of process control in modern continuous casting operations.

The present internship project focuses on analysing the influence of important process parameters on average heat flux using historical plant data obtained from the TSCR department of Tata Steel. Statistical analysis, Multiple Linear Regression (MLR), Random Forest analysis, and trial studies on mould and casting powder were carried out to identify the parameters that significantly affect heat transfer. The knowledge gained from these analyses is expected to assist in optimizing process conditions, improving shell formation, and reducing crack breakout during continuous casting.

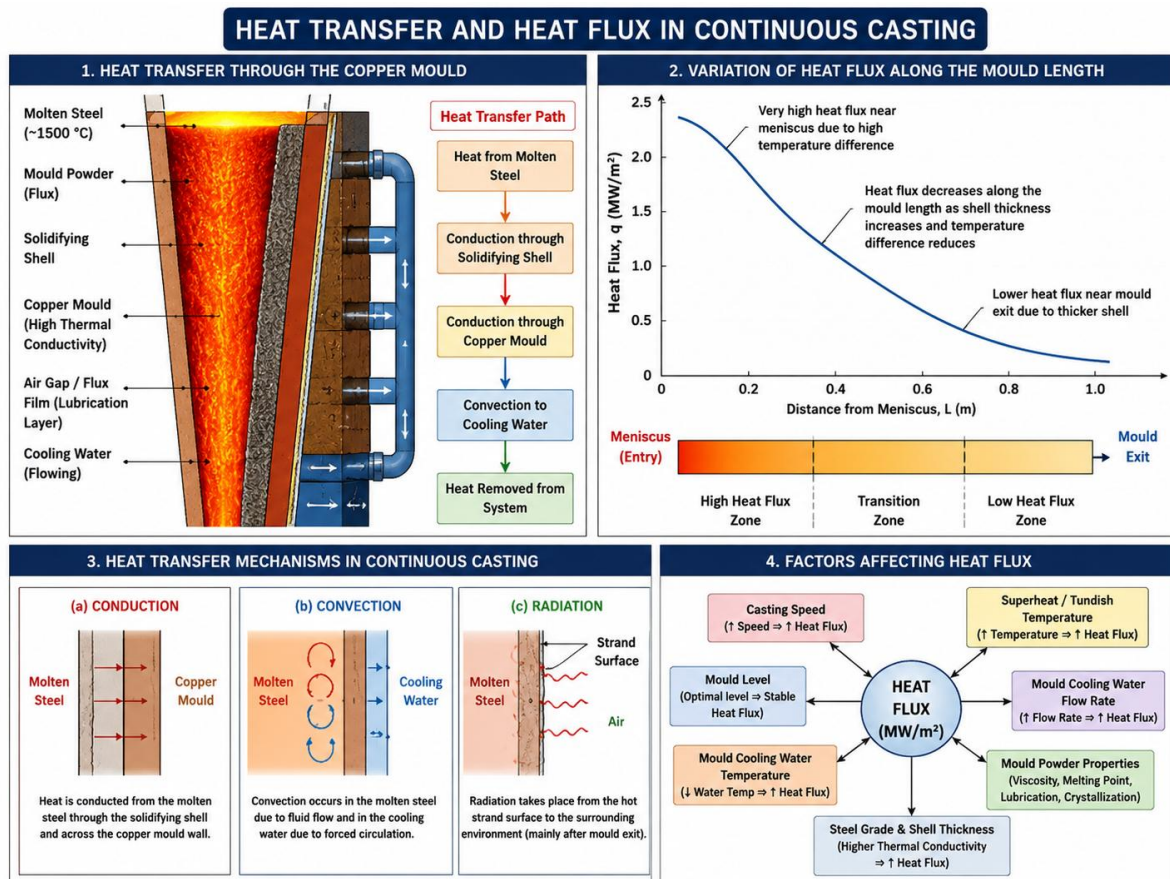


Fig 6. Heat Transfer Mechanisms and Factors Affecting Heat Flux in Continuous Casting

1.5 Crack Formation and Breakout in Continuous Casting

The quality of continuously cast steel is highly dependent on the formation of a sound and uniform solid shell during the initial stages of solidification. If the shell develops defects or fails to attain sufficient thickness before leaving the mould, the risk of crack formation and breakout increases significantly. These defects not only affect the surface quality of the slab but also lead to production losses, equipment damage, and increased maintenance costs.

Longitudinal cracks are among the most common surface defects observed in continuously cast slabs. These cracks generally develop due to non-uniform heat extraction, uneven shell growth, excessive thermal stresses, improper lubrication, or fluctuations in mould operating conditions. Once initiated, longitudinal cracks may propagate along the casting direction, resulting in poor surface quality and reduced mechanical properties of the final rolled product.

Another major concern in continuous casting is breakout, which is considered one of the most severe operational failures. Breakout occurs when the solidified shell surrounding the molten steel becomes too thin or weak to withstand the ferrostatic pressure of the liquid core. As a result, the shell ruptures and molten steel escapes from the strand, leading to equipment damage, production interruption, safety hazards, and substantial economic losses.

Several process parameters contribute to crack formation and breakout. These include casting speed, mould level fluctuations, excessive or insufficient heat flux, improper mould powder performance, high superheat, inadequate secondary cooling, mould oscillation abnormalities, and variations in steel composition. Since these parameters are interrelated, continuous monitoring and optimization are essential for maintaining stable casting conditions.

The mould plays a crucial role in determining the quality of the initial solid shell. Stable heat transfer between the molten steel and the water-cooled copper mould promotes uniform shell growth, whereas unstable heat transfer may cause localized shell thinning or sticking. Similarly, mould powder serves multiple functions, including lubrication, thermal insulation, and control of heat transfer. Improper mould powder behaviour can disturb lubrication and heat extraction, thereby increasing the probability of crack formation.

Modern steel plants employ various breakout prediction systems that continuously monitor process variables such as mould thermocouple temperatures, mould level, casting speed, heat flux, and shell behaviour. These systems generate alarms whenever abnormal thermal conditions are detected. However, accurate prediction still requires detailed analysis of historical plant data and an understanding of the relationship between process parameters and shell formation.

In the present study, historical operating data from the TSCR department of Tata Steel were analysed to identify the process variables that significantly influence average heat flux. Statistical analysis, Multiple Linear Regression (MLR), Random Forest analysis, and breakout data analysis were performed to investigate the relationship between operating conditions and crack breakout. Furthermore, trial studies involving modified mould conditions and casting

powder were conducted to evaluate their effect on heat transfer and slab quality. The outcomes of these analyses are expected to support the development of improved operating practices for minimizing breakout and enhancing casting stability.

Industrial Observation

During the plant visit, the breakout monitoring system installed in the TSCR department was observed. Engineers explained that mould thermocouple temperatures, mould level, casting speed, and other critical parameters are continuously monitored in real time. Any abnormal thermal behaviour indicating shell thinning or sticking is immediately detected by the system, allowing corrective action to be taken before molten steel escapes through the shell.

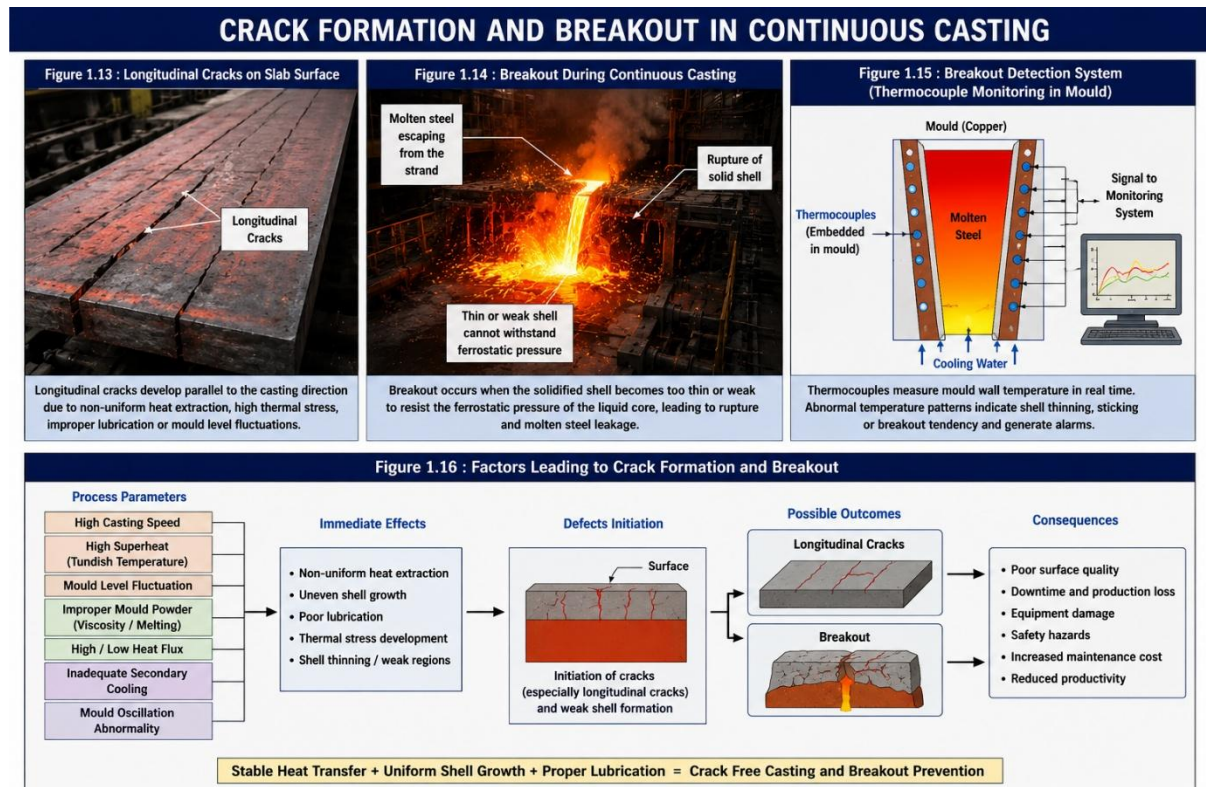


Fig 7. Crack Formation, Breakout Mechanism, and Breakout Detection System in Continuous Casting

1.6 Problem Statement

The Thin Slab Casting and Rolling (TSCR) process requires precise control of heat transfer during the initial stages of solidification to ensure the formation of a strong and uniform shell. Variations in process parameters such as casting speed, mould level, mould cooling conditions, and mould powder characteristics can significantly influence heat flux within the mould. Unstable heat transfer may result in non-uniform shell growth, longitudinal surface cracks, and, in severe cases, breakout, leading to production losses, equipment damage, increased maintenance costs, and safety hazards.

Although modern casting systems continuously monitor numerous process variables, identifying the parameters that have the greatest influence on heat flux remains a challenging task due to the complex interactions among process variables. Therefore, a systematic analysis of historical plant data is required to understand these relationships and identify the dominant factors governing heat transfer.

In addition, Tata Steel has undertaken trials involving modified mould conditions and casting powder to improve casting performance. Evaluating the effectiveness of these trials requires detailed statistical analysis and comparison with existing operating conditions. Hence, this project focuses on analysing plant data using statistical and machine learning techniques to identify the critical process parameters affecting heat flux and to support process optimization for reducing crack breakout in the TSCR process.

1.7 Objectives of the Study

The primary objectives of this internship project are:

- To study the Thin Slab Casting and Rolling (TSCR) process at Tata Steel.
- To understand the mechanism of heat transfer and heat flux during continuous casting.
- To perform an extensive literature review on heat flux, shell formation, crack generation, and breakout phenomena.
- To study the influence of important process parameters on average heat flux.
- To analyse historical plant data using statistical methods.
- To identify significant process parameters through scatter plot analysis and correlation analysis.
- To develop a Multiple Linear Regression (MLR) model for predicting average heat flux.
- To apply Random Forest analysis to determine the relative importance of different process parameters.

- To analyse breakout records for identifying process trends associated with breakout occurrence.
- To compare the performance of existing and trial mould/casting powder conditions.
- To understand the Design of Experiments (DOE) methodology adopted for future mould and casting powder optimization.

1.8 Scope of the Study

The scope of this work is limited to the analysis of process parameters influencing heat flux in the Thin Slab Casting and Rolling (TSCR) process at Tata Steel. The study includes literature review, plant visits, understanding of the continuous casting process, collection and preprocessing of historical plant data, statistical analysis, Multiple Linear Regression (MLR), Random Forest analysis, breakout data analysis, and evaluation of trial mould and casting powder performance. The study is intended to provide insights into process optimization and support future Design of Experiments (DOE) aimed at improving heat transfer and reducing crack breakout during continuous casting.

1.9 Organization of the Report

The report is organized into eight chapters.

Chapter 1 presents an introduction to Tata Steel, the continuous casting process, Thin Slab Casting and Rolling (TSCR), heat transfer, heat flux, crack formation, breakout, the problem statement, objectives, and the scope of the study.

Chapter 2 presents the literature review related to continuous casting, heat transfer mechanisms, shell formation, mould heat flux, mould powder behaviour, crack formation, breakout, and previous research findings.

Chapter 3 describes the TSCR plant, process flow, plant observations, data collection procedure, and methodology adopted during the internship.

Chapter 4 presents the preprocessing of plant data, exploratory data analysis, scatter plot analysis, correlation analysis, and interpretation of the observed relationships.

Chapter 5 discusses the application of Multiple Linear Regression (MLR), Random Forest analysis, and breakout data analysis for identifying the critical process parameters influencing heat flux.

Chapter 6 presents the comparative analysis of the trial mould and casting powder with the existing operating conditions and discusses the Design of Experiments (DOE) proposed for process optimization.

Chapter 7 summarizes the major findings of the study, presents the conclusions, and provides recommendations for future work.

Chapter 8 includes the references, appendices, additional plots, regression outputs, and supplementary information used during the project.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The literature review forms an essential part of this study by providing a comprehensive understanding of the Thin Slab Casting and Rolling (TSCR) process, heat transfer phenomena, solidification behaviour, mould heat flux, and crack breakout mechanisms during continuous casting. It also helps identify the important process parameters that influence casting stability and product quality.

Continuous casting has been extensively studied over the past several decades because of its significant contribution to improving steel production efficiency and product quality. Researchers have investigated various aspects of the process, including heat transfer in the mould, shell growth, mould powder behaviour, mould oscillation, secondary cooling, breakout prediction, and the application of computational as well as data-driven techniques for process optimization.

One of the primary challenges in continuous casting is maintaining stable heat transfer during the initial stages of solidification. Uniform heat extraction is necessary to develop a sufficiently strong solid shell capable of supporting the molten steel core throughout the casting process. Variations in process parameters may disturb this balance, leading to shell thinning, longitudinal surface cracks, sticker breakouts, and deterioration of slab quality.

Several researchers have reported that process variables such as casting speed, superheat, mould level, mould cooling water flow rate, mould cooling water temperature, casting powder properties, steel chemistry, and mould oscillation significantly influence mould heat flux. These variables interact in a complex manner, making it difficult to identify their individual contributions through conventional analytical methods alone.

In recent years, statistical analysis and machine learning techniques have become increasingly important tools for analyzing large volumes of historical plant data. Methods such as correlation analysis, Multiple Linear Regression (MLR), Random Forest, Support Vector Machines (SVM), Artificial Neural Networks (ANN), and other predictive models have been successfully applied to identify critical process parameters, predict casting behaviour, and improve operational decision-making in steel plants.

The present literature review summarizes the theoretical concepts and previous research findings relevant to heat transfer, mould heat flux, shell formation, crack generation, breakout phenomena, and process optimization. These studies provide the scientific foundation for the data analysis and experimental investigations carried out during this internship project at Tata Steel.

2.2 Heat Transfer During Continuous Casting

Heat transfer is one of the most critical aspects governing the quality of continuously cast steel slabs. During the continuous casting process, molten steel enters the water-cooled copper mould at a temperature of approximately 1550–1600°C. As the liquid steel comes into contact with the mould walls, rapid heat extraction begins, resulting in the formation of a thin solid shell around the liquid core.

Initially, the rate of heat transfer is very high because of the large temperature difference between the molten steel and the copper mould. As the shell thickness increases, the thermal resistance also increases, reducing the rate of heat extraction. The shell must become sufficiently thick before the strand exits the mould to withstand the ferrostatic pressure of the liquid steel inside.

The efficiency of heat transfer depends on several operating parameters, including casting speed, mould level, mould powder properties, cooling water flow rate, superheat of liquid steel, mould taper, and the contact between the shell and mould wall. Any disturbance in these parameters may result in uneven heat extraction, causing variations in shell thickness and increasing the possibility of crack formation.

In the secondary cooling zone, water sprays continue removing heat from the strand while allowing complete solidification before the slab enters the rolling stage. Uniform cooling throughout the casting process is essential to minimize thermal stresses and maintain product quality.

Since heat transfer directly controls shell growth and solidification behaviour, optimizing mould heat flux has become an important research area for reducing longitudinal surface cracks and breakout incidents in TSCR plants.

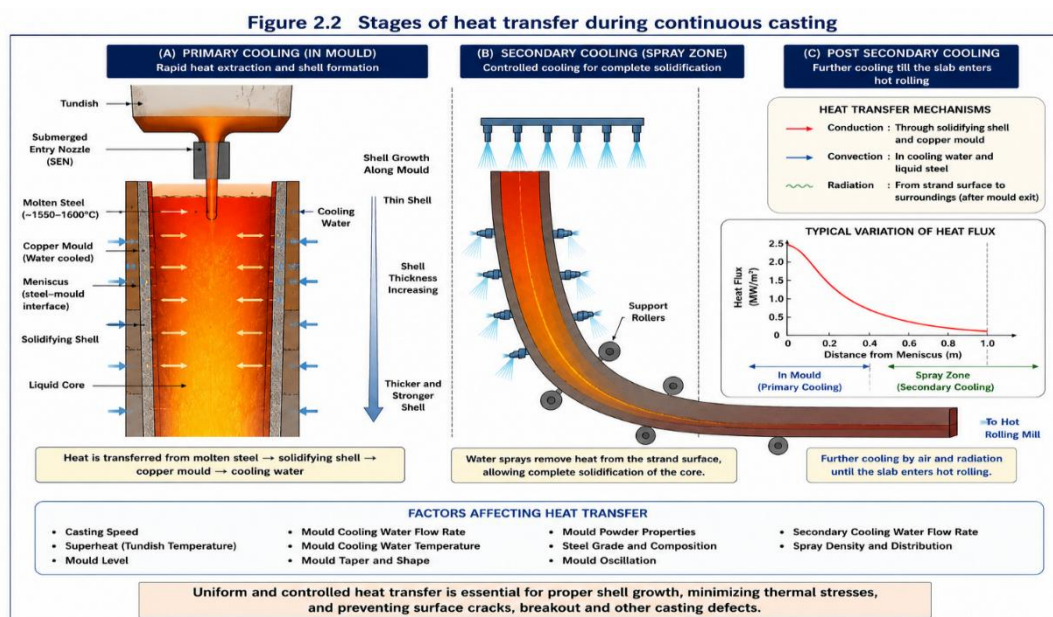


Fig 8. Crack Formation, Breakout Mechanism, and Breakout Detection System in Continuous Casting

Equation 2.1 – Heat Flux

$$q = \frac{Q}{A}$$

where:

- q = Heat flux (W/m²)
- Q = Heat transfer rate (W)
- A = Heat transfer area (m²)

Equation 2.2 – Fourier's Law of Heat Conduction

$$q = -k \frac{dT}{dx}$$

where:

- k = Thermal conductivity (W/m·K)
- $\frac{dT}{dx}$ = Temperature gradient

Equation 2.3 – Heat Removed by Cooling Water

$$Q = \dot{m}C_p(T_{out} - T_{in})$$

where:

- $\dot{m}$ = Mass flow rate of cooling water (kg/s)
- C_p = Specific heat of water
- T_{out} = Outlet water temperature
- T_{in} = Inlet water temperature

Figure 2.5 Factors Affecting Mould Heat Flux (Flowchart)

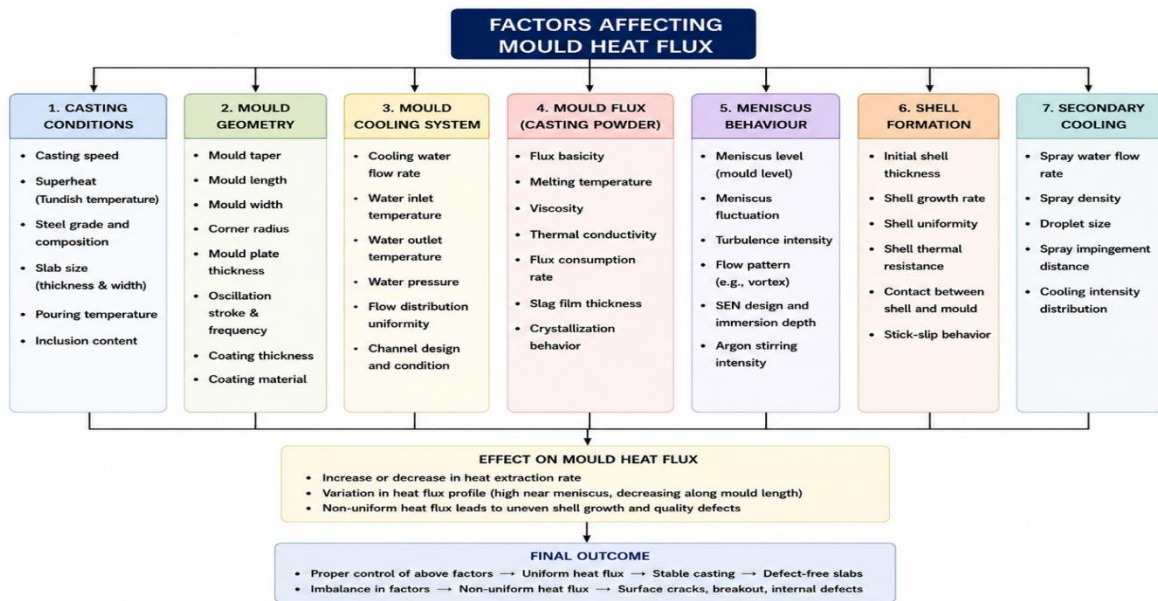


Figure 9. Flowchart Illustrating the Factors Affecting Mould Heat Flux in Continuous Casting

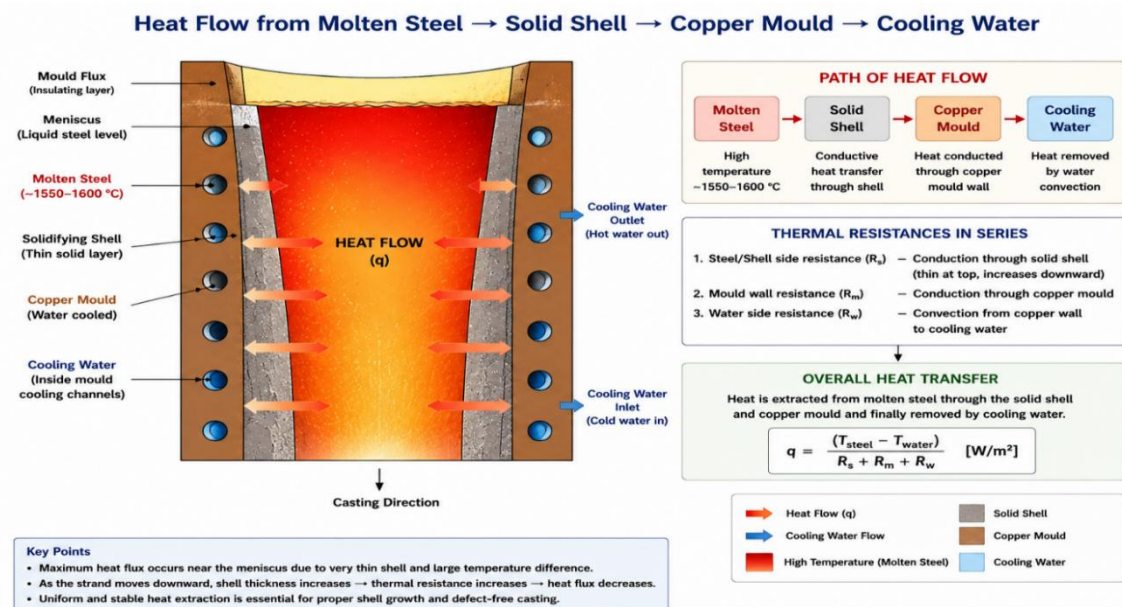


Fig 10: Heat Flow Path and Heat Transfer Mechanism in the Continuous Casting Mould

2.3 Mould Flux (Casting Powder)

Mould flux, commonly known as casting powder, is a specially formulated granular material added continuously onto the surface of molten steel inside the mould during continuous casting. As the powder comes into contact with the high-temperature molten steel, it undergoes a series of transformations from powder to liquid slag and finally forms a thin slag film between the copper mould and the solidifying steel shell. This slag film plays a crucial role in controlling

heat transfer, improving lubrication, preventing re-oxidation, and ensuring smooth withdrawal of the cast strand.

The performance of mould flux has a direct influence on shell formation and slab quality. An effective mould flux melts uniformly, infiltrates the mould-shell gap, and forms a stable liquid film that reduces friction between the moving shell and the copper mould. This lubrication minimizes the possibility of shell sticking and oscillation mark formation while allowing uniform heat extraction throughout the mould.

In addition to lubrication, mould flux acts as a thermal regulator. The thickness, viscosity, crystallization behaviour, and thermal conductivity of the slag film determine the amount of heat transferred from the solidifying shell to the mould. A highly conductive slag film increases heat extraction, whereas excessive crystallization reduces heat transfer by increasing thermal resistance. Therefore, selecting an appropriate mould flux is essential for maintaining optimum heat flux and stable shell growth.

Mould flux also provides thermal insulation at the meniscus by reducing heat loss from the exposed molten steel surface. Furthermore, it protects the liquid steel from atmospheric oxidation and absorbs non-metallic inclusions that rise to the steel surface, thereby improving steel cleanliness.

The chemical composition of mould flux generally consists of silica (SiO_2), calcium oxide (CaO), sodium oxide (Na_2O), calcium fluoride (CaF_2), alumina (Al_2O_3), magnesium oxide (MgO), and carbonaceous materials. The proportions of these constituents determine important physical properties such as melting temperature, viscosity, basicity, thermal conductivity, and crystallization tendency.

Several properties are considered while selecting mould flux for industrial casting operations:

- Melting temperature
- Viscosity
- Basicity
- Crystallization behaviour
- Thermal conductivity
- Carbon content
- Slag film thickness
- Powder consumption rate
- Lubrication efficiency
- Inclusion absorption capability

The selection of mould flux depends on steel grade, casting speed, slab dimensions, mould geometry, and desired surface quality. High-speed casting generally requires mould powders with rapid melting characteristics and stable lubrication to ensure uniform shell growth.

In the present internship project, casting powder is one of the major variables considered for process optimization. Tata Steel conducted industrial trials using a modified mould powder to investigate its influence on mould heat flux, shell formation, and crack breakout. The analysis and comparison of historical operating data with the trial data form an important part of this study and are expected to provide valuable insights into optimizing heat transfer during continuous casting.

Table 2.1 Typical Functions of Mould Flux

Function	Importance
Lubrication	Reduces friction between mould and shell
Heat transfer control	Maintains optimum heat flux
Thermal insulation	Prevents heat loss from meniscus
Inclusion absorption	Improves steel cleanliness
Oxidation prevention	Protects molten steel from atmosphere
Surface quality improvement	Reduces cracks and oscillation marks
Stable shell growth	Promotes uniform solidification

2.4 Mould Oscillation

Mould oscillation is an essential feature of the continuous casting process that facilitates the smooth withdrawal of the solidifying strand from the copper mould. Since the initial solid shell is in direct contact with the mould wall, there is a high possibility of sticking due to friction and thermal contraction. To prevent this, the mould is oscillated vertically with a controlled frequency and stroke throughout the casting operation.

The primary purpose of mould oscillation is to reduce friction between the mould wall and the solidifying shell. The periodic upward and downward motion of the mould allows molten mould flux to infiltrate the gap between the shell and the mould, forming a lubricating slag film. This lubrication minimizes shell sticking, ensures stable strand movement, and improves the surface quality of the cast slab.

Mould oscillation also influences heat transfer within the mould. Proper oscillation promotes uniform distribution of mould flux, resulting in stable heat extraction and uniform shell growth. However, improper oscillation parameters may disturb lubrication, causing uneven heat

transfer, localized shell thinning, oscillation marks, longitudinal surface cracks, and, in severe cases, breakout.

The oscillation cycle is characterized by two important parameters: **oscillation frequency** and **oscillation stroke**. The oscillation frequency represents the number of oscillation cycles per minute, whereas the oscillation stroke refers to the vertical displacement of the mould during each cycle. These parameters are optimized according to casting speed, steel grade, slab dimensions, and mould design to achieve efficient lubrication and stable casting conditions.

Another important concept associated with mould oscillation is **negative strip time**. During this period, the downward velocity of the mould becomes greater than the withdrawal speed of the strand, allowing mould flux to penetrate effectively into the mould-shell gap. Adequate negative strip time improves lubrication, reduces friction, and minimizes shell sticking. However, excessive negative strip time may increase oscillation mark depth and adversely affect slab surface quality.

The interaction between mould oscillation and mould flux is particularly important. A well-matched combination of oscillation parameters and mould powder properties ensures proper slag film formation, controlled heat transfer, and uniform shell growth. Therefore, optimization of both parameters is necessary to maintain stable heat flux and reduce the occurrence of longitudinal cracks and breakout.

Several researchers have reported that inappropriate oscillation conditions lead to deeper oscillation marks, unstable meniscus behaviour, increased thermal stress, and deterioration of slab quality. Modern TSCR plants continuously monitor oscillation parameters as part of the overall process control system to ensure safe and efficient operation.

Although mould oscillation was not directly analysed during the present internship, it is recognized as an important process variable that influences heat transfer and shell formation. Understanding its role provides a better interpretation of the relationship between mould heat flux, casting stability, and crack breakout.

Table 2.2 Important Oscillation Parameters

Parameter	Description
Oscillation Frequency	Number of oscillations per minute
Oscillation Stroke	Vertical displacement of the mould
Negative Strip Time	Period during which mould speed exceeds casting speed
Oscillation Mark	Surface impression produced due to mould movement
Lubrication	Formation of slag film between mould and shell

Figure 2.7 Working Principle of Mould Oscillation during Continuous Casting

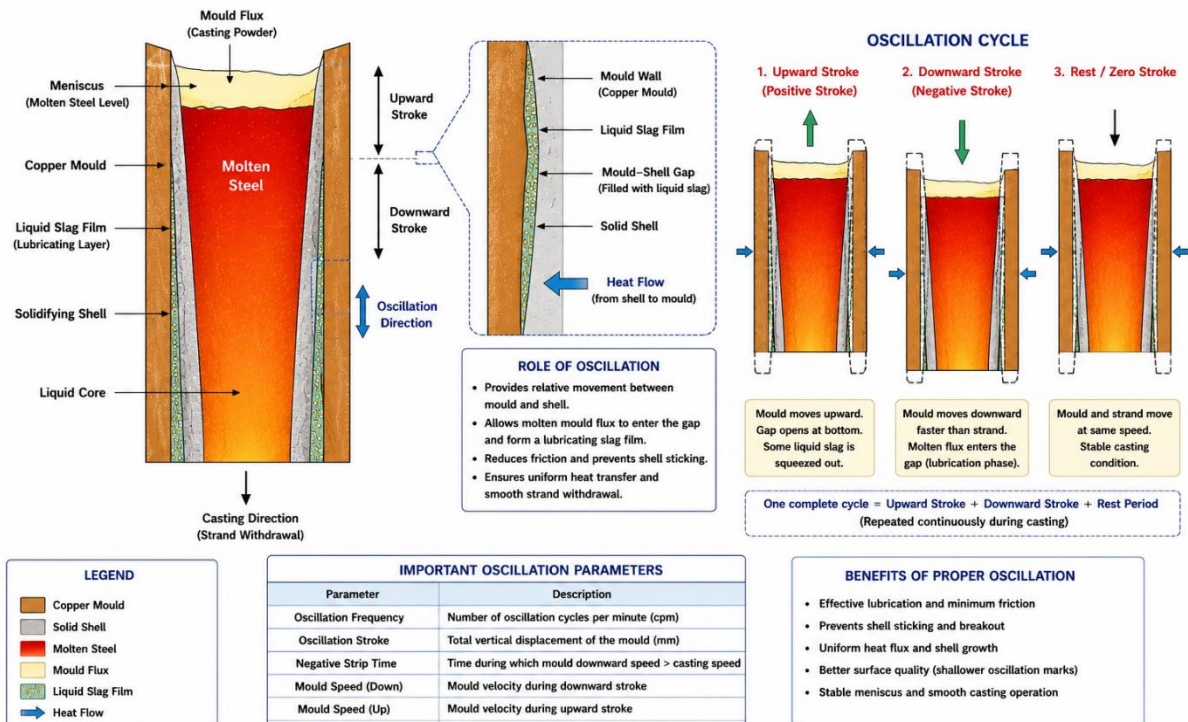


Figure 11: Working Principle of Mould Oscillation during Continuous Casting

2.5 Shell Formation During Continuous Casting

Shell formation is one of the most critical phenomena in the continuous casting process. It begins immediately after molten steel enters the water-cooled copper mould. Due to the large temperature difference between the molten steel and the mould wall, rapid heat extraction occurs, causing the outer layer of molten steel to solidify and form a thin solid shell. This shell surrounds the remaining liquid steel core and gradually thickens as the strand moves downward through the mould and secondary cooling zone.

The initial shell formed near the meniscus is extremely thin, typically only a few millimetres thick, and is therefore highly susceptible to deformation and cracking. As the strand progresses through the mould, continuous heat extraction causes the shell thickness to increase. The shell must attain sufficient strength before exiting the mould to withstand the ferrostatic pressure exerted by the liquid steel. If the shell remains too thin or develops weak regions, it may rupture, resulting in a breakout.

The growth of the solid shell is primarily controlled by heat transfer at the steel–mould interface. Uniform heat extraction promotes uniform shell growth, whereas localized variations in heat flux may produce regions of unequal shell thickness. Such non-uniform shell development is one of the major causes of longitudinal surface cracks, internal defects, and breakout.

Several process parameters influence shell formation, including casting speed, superheat, mould heat flux, mould level, mould oscillation, mould taper, mould flux properties, cooling

water flow rate, and secondary cooling intensity. Among these, heat flux plays a particularly significant role because it directly governs the rate of solidification. A stable and optimized heat flux produces a uniform shell with improved mechanical strength, while unstable heat transfer may result in thermal stresses and shell instability.

As solidification proceeds, the shell undergoes thermal contraction due to cooling. If the contraction is non-uniform or constrained by friction between the shell and the mould wall, tensile stresses may develop within the shell. These stresses can initiate surface cracks, especially near the corners and the funnel region of the mould where heat transfer conditions are more complex.

Researchers have shown that maintaining a balance between heat extraction and shell growth is essential for achieving high casting quality. Excessively rapid cooling may increase thermal stress, whereas insufficient cooling may produce inadequate shell thickness. Therefore, optimizing shell formation requires careful control of all process variables affecting heat transfer.

In the present study, shell formation is indirectly investigated through the analysis of average heat flux and associated process parameters. By identifying the variables that significantly influence heat transfer, the study aims to understand the factors responsible for uniform shell growth and to develop recommendations for minimizing crack breakout during the TSCR process at Tata Steel.

Equation 2.4 – Shell Growth Relationship

One of the commonly used empirical relationships for shell growth is:

$$S = k\sqrt{t}$$

where:

- **S** = Shell thickness (mm)
- **k** = Solidification constant (mm/ $\sqrt{s}$)
- **t** = Time (s)

This equation indicates that shell thickness increases approximately with the square root of solidification time.

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Process Parameter Effect on Shell Formation

Casting Speed	Higher speed reduces shell thickness
Heat Flux	Controls solidification rate

Process Parameter Effect on Shell Formation

Superheat	Higher superheat delays shell formation
Mould Flux	Controls lubrication and heat transfer
Mould Level	Influences meniscus stability
Cooling Water	Controls mould cooling efficiency
Secondary Cooling	Determines final shell thickness
Steel Grade	Affects solidification characteristics

2.6 Longitudinal Surface Crack Formation

Longitudinal surface cracks are among the most common and critical defects observed during the continuous casting of steel slabs. These cracks generally appear parallel to the casting direction and are formed due to non-uniform shell growth, excessive thermal stress, inadequate lubrication, and unstable heat transfer within the mould. If not detected and controlled at an early stage, these cracks may propagate during subsequent rolling operations or, in severe cases, develop into breakout incidents.

The initiation of longitudinal cracks begins during the early stages of solidification when the solid shell is still very thin and mechanically weak. Uneven heat extraction from the mould causes variations in shell thickness across the slab width. Regions experiencing excessive heat removal contract more rapidly than adjacent regions, producing tensile thermal stresses. When these stresses exceed the strength of the solidifying shell, surface cracks begin to develop.

The meniscus region is particularly susceptible to crack formation because shell formation starts at this location. Fluctuations in mould level, improper mould powder infiltration, unstable heat flux, and variations in casting speed can disturb shell growth, resulting in localized thinning and crack initiation. Similarly, the funnel region of the mould experiences complex heat transfer conditions due to changes in mould geometry, making it another critical location for longitudinal crack formation.

Several process variables influence the occurrence of longitudinal surface cracks. High casting speed reduces the residence time of the strand inside the mould, producing a thinner shell. Excessive superheat delays solidification and weakens shell formation. Improper mould powder properties affect lubrication and heat transfer, while poor mould oscillation conditions increase friction between the shell and the mould wall. Variations in cooling water flow rate, mould taper, steel chemistry, and secondary cooling conditions also contribute to crack formation.

Research has shown that maintaining stable and uniform heat flux is one of the most effective methods for minimizing longitudinal cracks. Uniform shell growth reduces thermal stress concentrations and improves the mechanical strength of the shell before it exits the mould.

Modern steel plants continuously monitor mould heat flux and process parameters to detect abnormal operating conditions and prevent defect formation.

In the present internship project, the relationship between process parameters and average heat flux was investigated using statistical and machine learning techniques. Since heat flux directly influences shell growth, identifying the most significant parameters provides valuable insights for minimizing longitudinal surface cracks and reducing the probability of breakout in the TSCR process.

Common Causes of Longitudinal Surface Cracks

Cause	Effect
Uneven heat transfer	Non-uniform shell growth
High casting speed	Thin shell formation
High superheat	Delayed solidification
Poor mould flux lubrication	Increased friction
Mould level fluctuation	Meniscus instability
Improper mould oscillation	Deep oscillation marks
Excessive thermal stress	Crack initiation
Non-uniform cooling	Shell distortion

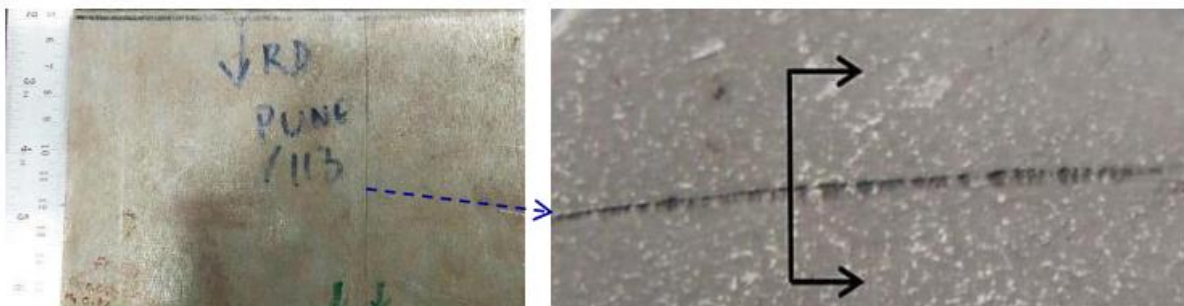


Figure 3: Longitudinal facial crack on one of the hot rolled low carbon steel sheets.

Figure 12: Longitudinal Surface Crack Observed on a Hot-Rolled Steel Sheet

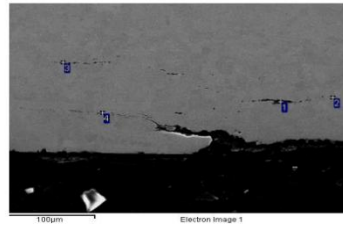


Fig.4: SEM image of LFC

Figure 13: SEM Micrograph Showing a Longitudinal Facial Crack (LFC)

2.7 Breakout Phenomenon in Continuous Casting

Breakout is one of the most severe operational failures encountered during the continuous casting of steel. It occurs when the solidified shell surrounding the liquid steel becomes too thin or weak to contain the molten core. As a result, the shell ruptures and molten steel escapes from the strand, causing severe equipment damage, production interruption, safety hazards, and significant economic losses. Because of these consequences, breakout prevention is considered one of the highest priorities in modern steel plants.

During continuous casting, a thin solid shell begins to form immediately after molten steel enters the water-cooled mould. This shell gradually thickens as heat is extracted through the mould and secondary cooling system. The shell must develop sufficient thickness and mechanical strength before leaving the mould to withstand the ferrostatic pressure of the liquid steel inside. If shell growth is insufficient or non-uniform, weak regions are formed, making the strand highly susceptible to rupture.

Several factors contribute to breakout. Uneven heat transfer is one of the primary causes, as it produces localized shell thinning and non-uniform solidification. High casting speed decreases the residence time of the strand inside the mould, reducing shell thickness. Excessive superheat delays the solidification process, while poor mould powder performance affects lubrication and heat transfer. Fluctuations in mould level, improper mould oscillation, nozzle clogging, cooling water abnormalities, and unstable casting conditions further increase the likelihood of breakout.



Figure 14: Hot Steel Slab Produced During the Continuous Casting Process

One of the most common types of breakout is the **sticker breakout**. This occurs when the solidifying shell sticks to the mould wall due to insufficient lubrication. As the strand continues to move downward, the stuck shell tears apart, producing an extremely thin region through which molten steel eventually escapes. Sticker breakout is often associated with abnormal heat flux patterns and sudden changes in mould thermocouple temperatures.

To minimize the risk of breakout, modern continuous casting machines are equipped with **Breakout Prediction Systems (BOPS)**. These systems continuously monitor mould thermocouple temperatures, heat flux, mould level, casting speed, cooling water conditions, and other process variables. Advanced algorithms analyse the thermal behaviour of the mould and generate alarms whenever abnormal conditions indicating shell sticking or shell thinning are detected. Operators can then reduce casting speed or stop casting before an actual breakout occurs.

In recent years, data-driven approaches such as statistical analysis, machine learning, and artificial intelligence have become increasingly important for breakout prediction. Historical plant data are analysed using techniques such as Multiple Linear Regression (MLR), Random Forest, Support Vector Machine (SVM), Artificial Neural Networks (ANN), and Logistic Regression to identify abnormal operating conditions and improve the accuracy of breakout prediction systems.

During this internship, historical plant data and breakout records from the TSCR department were studied to investigate the relationship between important process variables and heat flux. Random Forest analysis identified the most influential variables affecting heat transfer, while Multiple Linear Regression quantified the relationship between operating parameters and average heat flux. In addition, breakout records are being analysed to identify process trends associated with breakout occurrence and to support future optimization of mould design and casting powder through Design of Experiments (DOE).

Types of Breakout

Breakout Type	Description
Sticker Breakout	Shell sticks to mould due to poor lubrication and tears during withdrawal.
Thin Shell Breakout	Insufficient shell thickness caused by inadequate heat extraction.
Corner Breakout	Occurs due to excessive thermal stress and uneven cooling at slab corners.
Bulging Breakout	Shell deformation between support rollers leading to rupture.

Factors Responsible for Breakout

Parameter	Effect
High Casting Speed	Reduces shell thickness
Uneven Heat Flux	Causes localized shell thinning
High Superheat	Delays solidification
Poor Mould Flux	Reduces lubrication
Mould Level Fluctuation	Disturbs meniscus stability
Mould Oscillation Problems	Increases shell sticking
Cooling Water Variation	Produces non-uniform cooling
Nozzle Clogging	Alters steel flow pattern

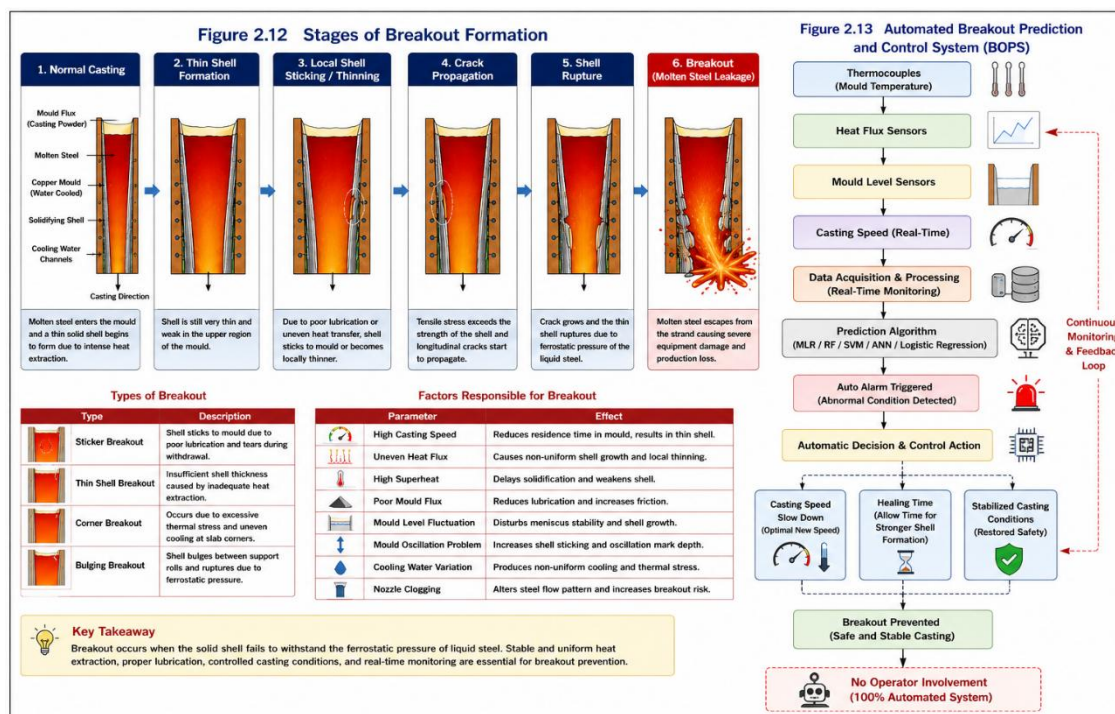


Figure 16: Stages of Breakout Formation and the Automated Breakout Prediction and Control System (BOPS)

2.8 Factors Affecting Heat Flux

Heat flux in the continuous casting mould is influenced by several interdependent process parameters. These parameters determine the rate of heat extraction from molten steel, directly affecting shell growth, surface quality, and breakout prevention. Since the continuous casting

process involves simultaneous heat transfer, fluid flow, and solidification, even small variations in operating conditions can significantly alter the heat flux profile.

Understanding the influence of these parameters is essential for maintaining stable casting conditions and minimizing defects. Researchers have reported that process variables such as casting speed, tundish temperature, mould level, mould flux properties, cooling water conditions, and mould geometry collectively determine the thermal behaviour inside the mould.

During this internship, several of these parameters were statistically analysed using historical plant data to determine their relationship with average heat flux. Correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis were performed to identify the most influential process variables. The major factors affecting mould heat flux are discussed below.

2.8.1 Casting Speed

Casting speed is one of the most influential operating parameters affecting heat transfer during continuous casting. Increasing the casting speed reduces the residence time of the strand inside the mould, resulting in a thinner solid shell at mould exit. Higher casting speeds generally increase instantaneous heat extraction near the meniscus but may reduce the overall shell thickness, thereby increasing the risk of shell instability and breakout. Therefore, an optimum casting speed must be maintained to balance productivity and casting safety.

2.8.2 Tundish Temperature (Superheat)

Tundish temperature represents the temperature of molten steel before entering the mould. Higher superheat delays the onset of solidification because additional heat must be removed before shell formation begins. Excessive superheat therefore reduces shell thickness, increases thermal stress, and raises the probability of crack formation. Conversely, excessively low superheat may reduce steel fluidity and create casting difficulties.

2.8.3 Mould Level

Maintaining a stable mould level is essential for achieving uniform shell growth. Fluctuations in mould level disturb the meniscus region, affecting mould flux infiltration and causing uneven heat extraction. Large mould level variations often result in localized shell thinning, slag entrapment, and surface defects.

2.8.4 Mould Flux Ratio

The properties and consumption rate of mould flux significantly influence lubrication and heat transfer. Proper mould flux infiltration produces a stable slag film between the mould and the

shell, ensuring smooth strand withdrawal and controlled heat extraction. Variations in mould flux characteristics alter thermal resistance and directly affect the measured heat flux.

2.8.5 Cooling Water Conditions

The mould cooling system continuously removes heat extracted from molten steel. Cooling water flow rate, inlet temperature, outlet temperature, and pressure determine the efficiency of heat removal. Variations in cooling water conditions may lead to unstable heat transfer and non-uniform shell growth.

2.8.6 Mould Geometry

The design of the copper mould, particularly the funnel mould used in TSCR, significantly influences heat transfer distribution. Mould taper, plate thickness, and cooling channel arrangement determine how uniformly heat is extracted from different regions of the strand. In the present study, future Design of Experiments (DOE) involves evaluation of modified funnel mould designs to improve heat transfer and reduce crack breakout.

2.8.7 Steel Grade

Different steel grades possess different thermal properties, chemical compositions, and solidification characteristics. These differences influence shell growth, heat transfer, and susceptibility to surface defects. Therefore, casting parameters are often optimized according to the steel grade being produced.

Table 2.3 Factors Affecting Heat Flux

Parameter	Influence on Heat Flux	Effect on Casting
Casting Speed	High	Controls shell thickness
Tundish Temperature	High	Delays solidification
Mould Level	Medium	Meniscus stability
Mould Flux	High	Lubrication & thermal resistance
Cooling Water Flow	High	Heat removal efficiency
Mould Geometry	Medium	Heat distribution
Steel Grade	Medium	Solidification characteristics

2.9 Previous Research Work

A considerable amount of research has been carried out to understand heat transfer, shell formation, crack generation, and breakout during continuous casting. Researchers have investigated the influence of process parameters, mould design, casting powder, and cooling conditions on slab quality using experimental studies, numerical simulations, statistical methods, and machine learning techniques. Some important contributions from previous studies are summarized below.

Research Work 1

Thomas et al. investigated the heat transfer mechanism during continuous casting and reported that the formation of a uniform solid shell is essential for preventing defects such as longitudinal cracks and breakout. Their study demonstrated that non-uniform heat extraction results in localized shell thinning, increasing the probability of shell failure during casting. They also emphasized the importance of optimizing mould cooling and maintaining stable operating conditions.

Research Work 2

Brimacombe and co-workers studied shell growth and thermal stress development during continuous casting. Their work showed that excessive thermal gradients generate tensile stresses within the solidifying shell, leading to crack initiation. The study concluded that controlling heat transfer and cooling conditions is essential for minimizing thermal stress and improving slab quality.

Research Work 3

Several researchers have investigated the influence of mould flux on heat transfer and lubrication. Their studies showed that mould powder performs multiple functions, including lubrication, thermal insulation, prevention of oxidation, and control of heat extraction. The viscosity, melting behaviour, and crystallization characteristics of mould flux significantly influence shell growth and surface quality.

Research Work 4

Researchers studying breakout prediction systems reported that mould thermocouple temperatures and heat flux measurements provide valuable information for detecting shell sticking and abnormal thermal conditions. Modern Breakout Prediction Systems (BOPS) continuously monitor mould temperature, casting speed, mould level, and cooling conditions to prevent breakout by generating early warning signals.

Research Work 5

Recent studies have applied machine learning techniques such as Multiple Linear Regression (MLR), Random Forest, Support Vector Machine (SVM), Artificial Neural Networks (ANN), and Logistic Regression for analysing historical plant data. These methods have been found effective in identifying important process variables, predicting heat transfer behaviour, and supporting process optimization in continuous casting operations.

Research Work 6

Research on funnel mould technology has demonstrated that mould geometry significantly influences heat transfer distribution and shell formation. Proper funnel mould design improves steel flow characteristics, promotes uniform shell growth, and reduces longitudinal crack formation. Continuous optimization of mould geometry remains an active area of research in Thin Slab Casting and Rolling (TSCR) plants.

Research Work 7

Industrial studies conducted at Tata Steel investigated the causes of longitudinal surface cracks in automotive-grade steel produced in the TSCR process. The researchers identified mould heat transfer, meniscus behaviour, mould powder performance, shell growth, and funnel mould characteristics as the major factors influencing crack formation. Based on plant investigations and metallurgical analysis, several operational improvements were recommended to reduce longitudinal surface cracks and improve product quality.

Summary of Previous Studies

The literature indicates that stable heat transfer is the key requirement for producing defect-free continuously cast slabs. Previous studies consistently highlight casting speed, superheat, mould level, mould flux behaviour, cooling conditions, and mould geometry as the dominant variables controlling shell formation and breakout. In recent years, machine learning and statistical analysis have emerged as powerful tools for analysing plant data and optimizing casting parameters. These findings provide the scientific basis for the present internship project, which focuses on identifying the process parameters affecting average heat flux and evaluating trial mould and casting powder conditions for reducing crack breakout in the TSCR process.

2.10 Research Gap

A comprehensive review of the available literature indicates that significant research has been conducted on heat transfer, shell formation, longitudinal crack formation, breakout mechanisms, mould flux behaviour, and continuous casting process optimization. Several researchers have employed experimental investigations, numerical simulations, water

modelling, and metallurgical studies to understand the influence of different process parameters on slab quality.

Recent studies have also demonstrated the usefulness of statistical methods and machine learning techniques for analysing large volumes of industrial process data. Methods such as Multiple Linear Regression (MLR), Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) have shown promising results in predicting casting behaviour and identifying the most influential operating parameters.

Although these studies provide valuable insights into the continuous casting process, most of them primarily focus on theoretical analysis, laboratory experiments, numerical modelling, or specific process variables. Limited research has been reported on the combined application of statistical analysis, machine learning, and industrial plant data for optimizing mould heat flux in an operating Thin Slab Casting and Rolling (TSCR) plant. Furthermore, very few studies have evaluated the influence of newly developed mould designs and casting powders using actual production trial data.

The present internship project addresses these gaps by analysing historical process data obtained from the TSCR department of Tata Steel. Correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis have been employed to identify the critical process parameters affecting average heat flux. In addition, industrial trial data involving a modified funnel mould and casting powder will be compared with the existing operating conditions to evaluate improvements in heat transfer behaviour. The findings of this work are expected to support future Design of Experiments (DOE) and contribute towards minimizing crack breakout, improving shell formation, and enhancing the overall stability of the continuous casting process.

CHAPTER 3

PLANT OVERVIEW AND METHODOLOGY

3.1 Introduction

This chapter presents an overview of Tata Steel, Jamshedpur, and the Thin Slab Casting and Rolling (TSCR) department where the internship was carried out. It describes the industrial process adopted for steel production, the major equipment involved in the TSCR process, and the observations made during plant visits. The chapter also outlines the methodology followed during the internship, including literature review, plant process understanding, data collection, preprocessing, statistical analysis, and machine learning techniques used for investigating the factors affecting average heat flux.

The internship was primarily focused on understanding the heat transfer behaviour during continuous casting and identifying the process parameters responsible for crack breakout in the TSCR process. A combination of industrial observations and historical plant data analysis was employed to achieve the objectives of the study.

The work commenced with an extensive literature review on continuous casting, shell formation, mould heat transfer, mould flux behaviour, longitudinal crack formation, and breakout mechanisms. This was followed by visits to the TSCR plant, where the complete steel casting process, mould cooling system, secondary cooling arrangement, and breakout prevention practices were studied in detail. Discussions with plant engineers provided valuable insights into the practical challenges associated with maintaining stable heat flux and ensuring defect-free slab production.

Subsequently, historical process data obtained from the TSCR department were analysed using statistical and machine learning techniques. Data preprocessing, scatter plot analysis, correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis were performed to determine the relationship between average heat flux and important process parameters. Industrial trials involving modified mould conditions and casting powder were also initiated, and the resulting data will be compared with the existing operating conditions to evaluate their influence on heat transfer and casting stability.

The methodology adopted during this internship integrates theoretical understanding with industrial practice and data-driven analysis. The outcomes of this work are expected to contribute towards improving mould heat transfer, minimizing crack breakout, and supporting future Design of Experiments (DOE) for process optimization in the TSCR plant.

3.1.1 Methodology Adopted During the Internship

The internship work was carried out in a systematic manner to understand the factors influencing heat flux and crack breakout in the Thin Slab Casting and Rolling (TSCR) process at Tata Steel. The study commenced with an extensive literature review of books, research papers, and internal Tata Steel technical reports related to continuous casting, heat transfer, shell formation, mould flux behaviour, longitudinal surface cracks, and breakout phenomena. This theoretical study provided a strong foundation for understanding the mechanisms governing heat transfer during continuous casting.

Following the literature review, detailed plant visits were conducted to gain practical exposure to the TSCR process. The complete production route, starting from the Steel Melting Shop (SMS) to the rolling mill, was studied. Important equipment such as the ladle, tundish, submerged entry nozzle (SEN), copper mould, secondary cooling system, support rollers, and breakout detection system were observed, and their functions were discussed with plant engineers.

Subsequently, historical process data were collected from the TSCR department. The data were preprocessed by removing incomplete records, handling missing values, and calculating average values of important process parameters wherever required. Exploratory Data Analysis (EDA) was then performed using scatter plots and descriptive statistics to understand the behaviour of different process variables.

To identify the relationship between average heat flux and various operating parameters, correlation analysis and Multiple Linear Regression (MLR) were carried out. Random Forest analysis was also employed to determine the relative importance of different process variables influencing heat flux. In addition, breakout records and industrial trials involving modified funnel moulds and casting powders were considered for further evaluation. Finally, the observations obtained from statistical analysis, machine learning techniques, and industrial trials were interpreted to develop recommendations for improving heat transfer and minimizing crack breakout in the TSCR process.

The methodology adopted in this study combines theoretical knowledge, industrial observations, statistical techniques, and machine learning approaches to provide a comprehensive understanding of the continuous casting process. By integrating literature findings with real plant data and practical insights gained during the internship, the study establishes a systematic framework for evaluating the influence of process parameters on heat flux. The results obtained from these analyses are further used to compare existing operating conditions with trial mould and casting powder configurations, thereby supporting future process optimization through Design of Experiments (DOE). The detailed methodology and corresponding results are presented in the subsequent chapters of this report.

3.2 Tata Steel Jamshedpur Works

Tata Steel Jamshedpur Works is India's first integrated steel manufacturing plant and one of the most advanced steel production facilities in the country. The plant consists of several interconnected production units that operate in a continuous manner to convert raw materials into high-quality finished steel products. The major units include Raw Material Handling, Coke Ovens, Sinter Plant, Blast Furnace, Steel Melting Shop (SMS), Thin Slab Casting and Rolling (TSCR), Hot Strip Mill, Cold Rolling Mill, Galvanizing Lines, and By-Product Management facilities.

The TSCR department plays a vital role in the production of high-quality hot rolled steel by integrating continuous casting and hot rolling into a single production line. This technology significantly reduces production time, energy consumption, and operational costs while improving product quality and manufacturing efficiency. During the internship, detailed observations were made regarding the functioning of the TSCR plant, the sequence of operations, and the interaction between various process parameters affecting continuous casting.

The steel manufacturing process begins in the Steel Melting Shop (SMS), where molten steel is produced and transferred through ladles to the continuous casting machine. The molten steel passes through the tundish and Submerged Entry Nozzle (SEN) before entering the water-cooled copper mould, where the initial solid shell is formed. The partially solidified strand then moves through the secondary cooling zone, support rollers, straightening section, and cutting unit before being transferred to the rolling mill for further processing.



Figure 3.1: Tata Steel Plant, Jamshedpur

During the plant visit, emphasis was placed on understanding the mould cooling system, heat transfer mechanisms, breakout prevention practices, mould powder behaviour, and process monitoring techniques adopted in the TSCR department. Discussions with plant engineers

provided practical knowledge regarding the influence of casting speed, tundish temperature, mould level, cooling water conditions, and mould flux on shell formation and slab quality.

The internship also provided exposure to modern digital process monitoring systems used in the TSCR plant. Various process parameters are continuously recorded through sensors and automation systems, enabling real-time monitoring of casting conditions. These data are utilized for process control, quality improvement, breakout prediction, and continuous optimization of plant operations. The historical datasets collected from these monitoring systems formed the basis for the statistical and machine learning analyses performed during this internship.

3.3 TSCR Process Flow

The Thin Slab Casting and Rolling (TSCR) process is an advanced steel manufacturing technology that integrates continuous casting and hot rolling into a single production line. Unlike conventional steel plants, where slabs are cast, cooled, stored, and reheated before rolling, the TSCR process directly transfers the continuously cast slab to the rolling mill while it is still at a high temperature. This significantly reduces energy consumption, production time, and operational costs while improving product quality.

The process begins at the Steel Melting Shop (SMS), where molten steel is produced and refined to the required chemical composition. The molten steel is transported to the casting machine using a ladle and is poured into the tundish, which acts as an intermediate reservoir to ensure a continuous and uniform flow of steel. From the tundish, the molten steel enters the copper mould through the Submerged Entry Nozzle (SEN). Inside the water-cooled mould, rapid heat extraction causes the formation of an initial solid shell around the liquid steel core.

After leaving the mould, the strand passes through the secondary cooling zone, where water sprays continue the solidification process while support rollers guide the strand. The strand is then straightened and transferred directly to the rolling mill without significant cooling. Finally, the rolled strip is cooled, coiled, and dispatched for further processing or delivery to customers. Throughout the process, various sensors continuously monitor casting speed, mould level, heat flux, cooling water conditions, and other operating parameters to ensure stable operation and high product quality.

3.4 Plant Observations During Internship

During the internship at Tata Steel, regular visits were made to the Thin Slab Casting and Rolling (TSCR) department to gain practical understanding of the continuous casting process. The complete production route, from molten steel transfer to hot rolling, was observed under the guidance of experienced plant engineers. These visits helped bridge the gap between theoretical knowledge and industrial practice.

One of the key observations was the operation of the water-cooled copper mould, where the initial solid shell of the slab is formed. It was observed that maintaining stable heat transfer inside the mould is essential for uniform shell growth and preventing surface defects. The

mould cooling system, mould level control, and casting speed were continuously monitored through automated control systems to ensure safe casting operations.

The behaviour of mould flux (casting powder) during casting was also studied. It was observed that the mould powder melts to form a lubricating slag film between the mould wall and the solidifying shell. This slag film not only reduces friction during strand withdrawal but also controls heat transfer and protects the molten steel from atmospheric oxidation.

The secondary cooling zone, consisting of multiple water spray nozzles and support rollers, was examined to understand its role in completing solidification. Uniform water spraying and proper roller alignment were found to be essential for maintaining strand stability and minimizing thermal stresses during cooling.

The plant engineers also explained the occurrence of longitudinal surface cracks and breakout during continuous casting. Discussions revealed that abnormal heat transfer, mould level fluctuations, improper lubrication, and excessive casting speed are among the major causes of shell thinning and breakout. Various monitoring systems, including thermocouples and process sensors, are used to detect abnormal casting conditions and prevent operational failures.

During the internship, the importance of data acquisition systems was also understood. Process parameters such as casting speed, tundish temperature, mould level, mould heat flux, cooling water conditions, and mould powder characteristics are continuously recorded and stored in the plant database. These historical datasets formed the basis for the statistical analysis and machine learning models developed during this project.

The plant visit also provided an opportunity to observe ongoing improvement activities within the TSCR department. Discussions with the project guide highlighted the industrial trials being conducted on modified funnel mould designs and casting powders to optimize heat transfer and reduce crack breakout. These observations directly supported the objectives of the present internship project.

3.5 Data Collection

Historical process data were collected from the Thin Slab Casting and Rolling (TSCR) department of Tata Steel for the present study. The dataset comprised plant operating data for the **month of April**, recorded during normal continuous casting operations. These data were automatically acquired through the plant's process monitoring and data acquisition systems, ensuring reliable measurements under actual industrial operating conditions.

The collected dataset contained numerous process parameters associated with continuous casting, including casting conditions, mould cooling system performance, heat transfer characteristics, and process stability. Based on the objectives of the internship and discussions with the project guide, only the parameters directly related to mould heat transfer and average heat flux were selected for detailed analysis.

The response variable considered in this study was **Average Heat Flux**, while the major input variables included **Average Casting Speed (AVG_CAST_SPEED)**, **Tundish Temperature (T_PUT)**, **Flux Ratio (FLUXRATIO_R)**, and **Mould Level (MOLD_LEV)**. These

parameters were selected because of their significant influence on shell formation, heat transfer, and breakout during continuous casting.

The collected plant data served as the foundation for subsequent analyses, including data preprocessing, exploratory data analysis, correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis. In addition, breakout records and industrial trial data for the modified funnel mould and casting powder were planned to be analysed for further process optimization as and when the data became available.

3.6 Data Preprocessing

Before performing statistical and machine learning analyses, the collected plant data were preprocessed to improve their quality and suitability for analysis. Data preprocessing is an essential step as raw industrial datasets often contain missing values, inconsistent records, and unnecessary variables that may affect the accuracy of the results.

Initially, the dataset was examined to identify missing or incomplete records. Rows containing insufficient or inconsistent information were removed to ensure data reliability. The relevant process parameters required for the study were then selected based on literature review, industrial observations, and discussions with the project guide.

Since the plant database contained a large number of variables, only those parameters directly related to mould heat transfer and average heat flux were retained for detailed investigation. The selected parameters included Average Heat Flux, Average Casting Speed (AVG_CAST_SPEED), Tundish Temperature (T_PUT), Flux Ratio (FLUXRATIO_R), and Mould Level (MOLD_LEV). After preprocessing, the cleaned dataset was imported into Microsoft Excel and Python for further analysis. Exploratory Data Analysis (EDA) was first performed using scatter plots and descriptive statistics to understand the behaviour of the selected variables. The processed dataset was then used for correlation analysis, Multiple Linear Regression (MLR), and Random Forest analysis to identify the process parameters having the greatest influence on average heat flux.

Table 3.3 Data Preprocessing Steps

Step	Description
Data Collection	April month plant data obtained from TSCR
Data Inspection	Checked for missing and inconsistent values
Data Cleaning	Removed incomplete records
Feature Selection	Selected relevant process parameters
Data Analysis	Imported into Excel and Python
Final Dataset	Used for Correlation, MLR and Random Forest

CHAPTER 4

DATA ANALYSIS AND RESULTS

4.1 Introduction

The primary objective of this study was to investigate the influence of important process parameters on Average Heat Flux during the Thin Slab Casting and Rolling (TSCR) process. Historical plant data collected from the TSCR department were analysed using statistical techniques to identify the process variables that significantly affect heat transfer within the mould.

The analysis was carried out in a systematic manner. Initially, Exploratory Data Analysis (EDA) was performed to understand the distribution and behaviour of the selected process parameters. Correlation analysis was then conducted to determine the strength and direction of the relationship between Average Heat Flux and the independent variables.

To quantify the combined influence of multiple process parameters, Multiple Linear Regression (MLR) analysis was performed. In addition, a Random Forest Regression model was developed to evaluate the relative importance of the selected variables in predicting Average Heat Flux. The results obtained from these analyses were interpreted with reference to the physical principles of heat transfer in continuous casting to identify the dominant process parameters affecting mould heat transfer.

	A	B	C	D	E	F
1		AVG	AVG_CAST_SPEED	T_PUT	FLUXRATIO_R	MOLD_LEV
2	AVG	1				
3	AVG_CAST	0.738296548	1			
4	T_PUT	0.715847696	0.636801138	1		
5	FLUXRATIO	0.479411987	0.303932622	0.2176538	1	
6	MOLD_LEV	-0.326962781	-0.262405575	-0.292518	-0.20655197	1

Figure 4.1: Correlation Matrix Between Average Heat Flux and Process Parameters

4.2 Correlation Analysis

Correlation analysis was performed to examine the relationship between Average Heat Flux (AVG) and the selected process parameters, namely Average Casting Speed (AVG_CAST_SPEED), Mould Level (MOLD_LEV), and Superheat. The correlation matrix is presented in Figure 4.1.

The results indicate that Average Casting Speed has the strongest positive correlation with Average Heat Flux ($r = 0.738$). This suggests that increasing the casting speed generally

increases the heat extraction rate during continuous casting because the mould removes heat more rapidly to form the required solid shell.

Mould Level exhibits a moderate negative correlation with Average Heat Flux ($r = -0.327$). An increase in mould level generally results in a thicker slag layer between the molten steel and the copper mould. Since the slag layer acts as an additional thermal resistance, heat transfer from the molten steel to the mould decreases, leading to a reduction in average heat flux. However, the correlation is moderate, indicating that mould level alone does not determine heat transfer and should be considered together with other operating parameters.

Superheat shows a very weak negative correlation with Average Heat Flux ($r = -0.073$). This indicates that under the analysed operating conditions, variations in superheat have only a small direct linear influence on heat transfer. Although superheat affects the thermal state of molten steel, its effect on average heat flux is comparatively smaller than that of casting speed and mould level.

The correlation among the independent variables was also examined. Average Casting Speed and Mould Level showed only weak correlation with each other, while Superheat exhibited negligible correlation with the remaining variables. The absence of strong correlations among the independent variables indicates that multicollinearity is not severe, making these variables suitable for Multiple Linear Regression analysis.

Overall, the correlation analysis identifies Average Casting Speed as the dominant parameter influencing Average Heat Flux, followed by Mould Level, whereas Superheat has the least direct influence among the selected variables.

4.3 Multiple Linear Regression (MLR)

Multiple Linear Regression (MLR) was employed to establish the relationship between Average Heat Flux and the selected process parameters. Unlike simple linear regression, MLR evaluates the combined influence of multiple independent variables on the response variable, thereby providing a more comprehensive understanding of the process.

For the present study, **Average Heat Flux (AVG)** was considered as the dependent variable, while **Average Casting Speed (AVG_CAST_SPEED)**, **Mould Level (MOLD_LEV)**, and **Superheat** were selected as independent variables. The regression model was developed using Microsoft Excel Data Analysis ToolPak.

The general form of the Multiple Linear Regression equation is

$$AVG = \beta_0 + \beta_1(AVG_CAST_SPEED) + \beta_2(MOLD_LEV) + \beta_3(SUPERHEAT)$$

where

AVG = Average Heat Flux

β_0 = Intercept

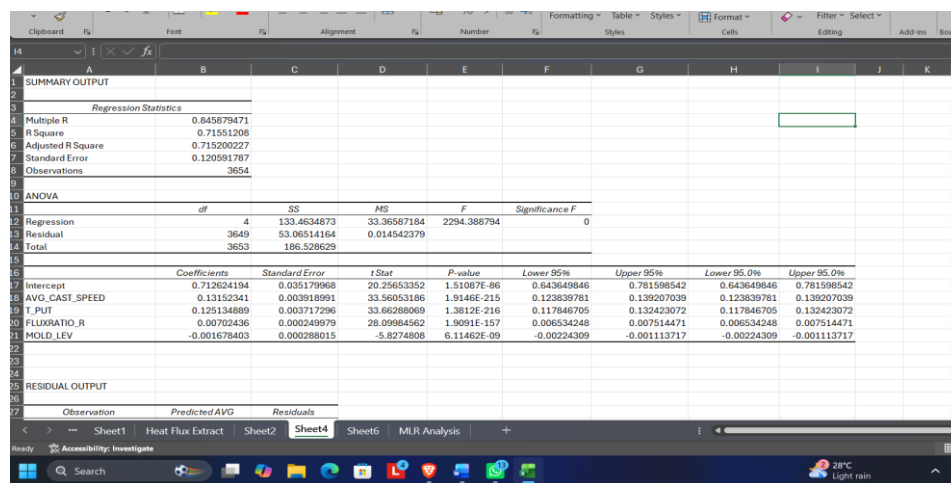
β_1 – β_3 = Regression coefficients

The regression coefficients indicate the direction and magnitude of the influence of each predictor on Average Heat Flux. A positive coefficient indicates that an increase in the corresponding parameter increases the predicted heat flux, whereas a negative coefficient indicates an inverse relationship.

The performance of the regression model is evaluated using statistical indicators such as the coefficient of determination (R^2), adjusted R^2 , F-statistic, p-values, and ANOVA.

4.4 Multiple Linear Regression Analysis

Multiple Linear Regression analysis was carried out to quantify the influence of Average Casting Speed, Mould Level, and Superheat on Average Heat Flux. The regression summary obtained from Microsoft Excel is presented in Figure 4.2.



SUMMARY OUTPUT									
Regression Statistics									
Multiple R	0.845679471								
R Square	0.71551208								
Adjusted R Square	0.715200227								
Standard Error	0.120591787								
Observations	3654								
ANOVA									
	df	SS	MS	F	Significance F				
Regression	4	133.4634873	33.36587184	2294.388794	0				
Residual	3649	53.06514164	0.014542379						
Total	3653	186.528629							
Coefficients									
		Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	0.712624194	0.035179968	20.25653352	1.51087E-86	0.643649846	0.781598542	0.643649846	0.781598542	
AVG_CAST_SPEED	0.13152341	0.003918991	33.56053186	1.9146E-215	0.123839781	0.139207039	0.123839781	0.139207039	
T_PUT	0.125134889	0.003717296	33.66288069	1.3812E-216	0.117846705	0.132423072	0.117846705	0.132423072	
FLUXRATIO_R	0.00702436	0.000249979	28.09984562	1.3991E-157	0.006534248	0.007514471	0.006534248	0.007514471	
MOLD_LEV	-0.001678403	0.000288015	-5.8274808	6.11462E-09	-0.00224309	-0.00113717	-0.00224309	-0.00113717	

Figure 4.2: Multiple Linear Regression Summary Output for the Existing Process Dataset

The regression model produced a **Multiple R value of 0.755**, indicating a good correlation between the observed and predicted values of Average Heat Flux. The coefficient of determination ($R^2 = 0.570$) indicates that approximately **57% of the variation in Average Heat Flux** is explained by the selected process parameters. The Adjusted R^2 value of **0.569** further confirms the reliability of the model.

The ANOVA results indicate that the regression model is statistically significant with an **F-statistic of 1602.73** and a **Significance F value close to zero**, confirming that the selected process parameters collectively have a significant influence on Average Heat Flux.

The regression coefficients reveal that **Average Casting Speed** has a positive coefficient, indicating that increasing casting speed increases Average Heat Flux. **Mould Level** has a negative coefficient, suggesting that higher mould levels reduce Average Heat Flux due to increased thermal resistance of the slag layer. Although **Superheat** has a negative coefficient, its p-value (**0.126**) is greater than 0.05, indicating that its influence is not statistically significant for the present dataset.

Overall, the Multiple Linear Regression analysis identifies **Average Casting Speed** as the dominant parameter governing Average Heat Flux, while Mould Level has a moderate inverse influence and Superheat has only a weak contribution.

Table 4.2 Regression Model Summary

Parameter	Value
Multiple R	0.755
R Square	0.570
Adjusted R Square	0.569
Standard Error	0.148
Observations	3635

4.5 Random Forest Analysis

Random Forest analysis was carried out using the selected process parameters to determine their relative importance in predicting Average Heat Flux.

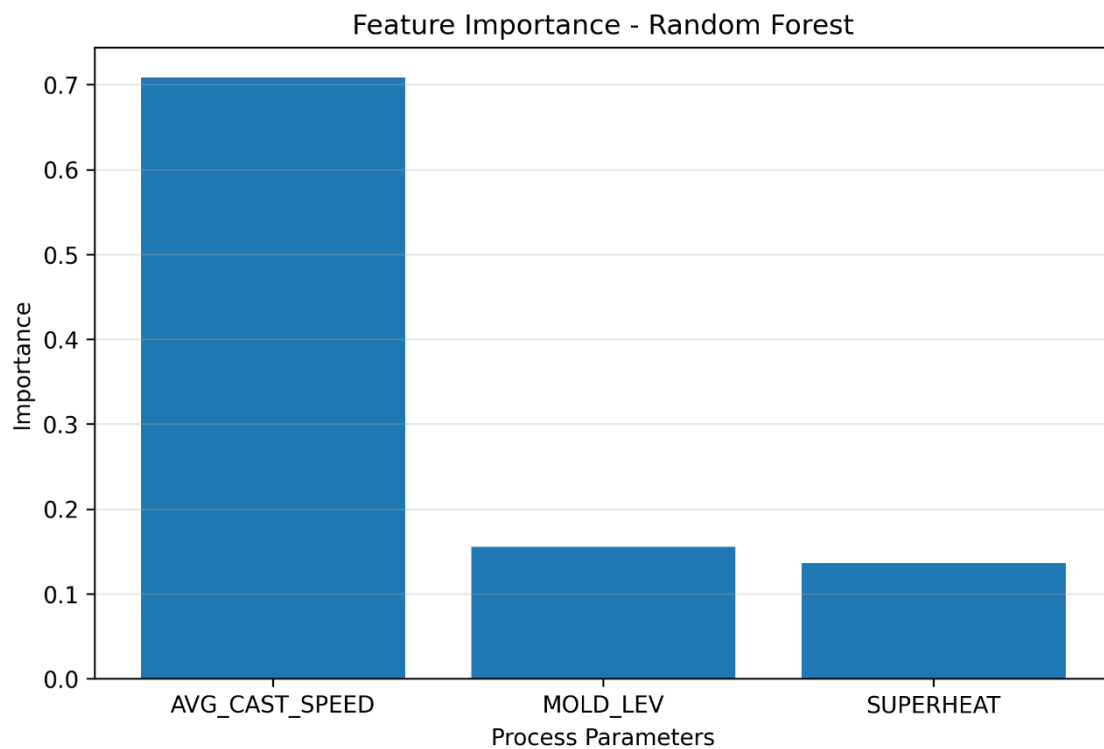


Figure 4.3: Feature Importance from Random Forest Analysis (Existing Process Dataset)

The Random Forest model indicates that **Average Casting Speed** is the most influential process parameter, contributing approximately **70.8%** of the total feature importance. This demonstrates that variations in casting speed have the greatest effect on mould heat transfer during continuous casting.

Mould Level contributes approximately **15.5%**, indicating a moderate influence on Average Heat Flux. Maintaining an appropriate mould level is therefore important for ensuring stable heat transfer and preventing casting defects.

Superheat contributes approximately **13.7%** of the total feature importance. Although it affects the thermal condition of molten steel entering the mould, its influence is comparatively smaller than that of casting speed and mould level.

The ranking obtained from the Random Forest model agrees well with the findings of the correlation analysis and Multiple Linear Regression model. All three analyses consistently identify **Average Casting Speed** as the dominant parameter influencing Average Heat Flux.

CHAPTER 5

ANALYSIS OF INDUSTRIAL TRIAL DATA

5.1 Introduction

Following the analysis of the historical operating dataset, an industrial trial was conducted in the Thin Slab Casting and Rolling (TSCR) plant to evaluate the influence of modified operating conditions on mould heat transfer behaviour. The objective of the trial was to analyse the effect of key process parameters on Average Heat Flux (AVG_HF) under actual production conditions and compare the obtained results with those of the existing process.

The trial data consisted of real-time operating measurements collected from the plant during the modified casting operation. Similar to the previous analysis, statistical techniques and machine learning methods were employed to identify the parameters having the greatest influence on mould heat transfer. The analysis included Correlation Analysis, Multiple Linear Regression (MLR), and Random Forest Regression.

The findings obtained from the industrial trial provide valuable insight into the thermal behaviour of the mould under modified operating conditions and help assess the effectiveness of process control strategies for improving casting stability and reducing breakout occurrence.

5.2 Data Preparation

Before performing the statistical analysis, the trial dataset was preprocessed to ensure data quality and consistency. Invalid and missing observations were removed, and the relevant process parameters associated with mould heat transfer were selected for further investigation.

The mould heat flux was measured individually at four locations of the mould, namely **HF_RIGHT**, **HF_LEFT**, **HF_FIX**, and **HF_LOOSE**. To obtain a single representative response variable, the **Average Heat Flux (AVG_HF)** was calculated as the arithmetic mean of these four heat flux measurements.

The process parameters selected for the trial data analysis were:

AVG_HF – Average Heat Flux (Response Variable)

CAST_SPEED_ACT – Actual Casting Speed

SUPERHEAT – Superheat of molten steel

MLF – Mould Level Fluctuation (deviation of mould steel level from its set point)

These variables were selected because they directly influence the heat transfer behaviour during continuous casting and were available throughout the trial period.

Table 5.1 Selected Process Parameters for Trial Analysis

Parameter	Description
AVG_HF	Average Heat Flux (Response Variable)
CAST_SPEED_ACT	Actual Casting Speed
SUPERHEAT	Superheat of Molten Steel
MLF	Mould Level Fluctuation

5.3 Correlation Analysis of Trial Data

Correlation analysis was performed to examine the linear relationship between **Average Heat Flux (AVG_HF)** and the selected process parameters under industrial trial conditions. The parameters considered in the analysis were **Actual Casting Speed (CAST_SPEED_ACT)**, **Superheat (SUPERHEAT)**, and **Mould Level Fluctuation (MLF)**.

The correlation matrix indicates that **Actual Casting Speed** exhibits the strongest positive correlation with Average Heat Flux, with a correlation coefficient of **0.726**. This indicates that increasing the casting speed increases the rate of heat extraction in the mould, resulting in higher Average Heat Flux during continuous casting.

MLF shows a weak positive correlation with Average Heat Flux ($r = 0.104$). Although the correlation is comparatively small, it suggests that mould level fluctuations still influence heat transfer behaviour. Since mould level stability affects the contact conditions between the solidifying shell and the mould, even small fluctuations can contribute to variations in heat extraction.

In contrast, **Superheat** exhibits a weak negative correlation ($r = -0.081$) with Average Heat Flux. This indicates that, within the analysed trial dataset, variations in superheat have only a limited inverse influence on the measured heat flux. The small magnitude of the correlation suggests that superheat was relatively well controlled during the industrial trial.

Overall, the correlation analysis indicates that **Actual Casting Speed** is the dominant parameter influencing Average Heat Flux in the trial dataset, while **MLF** has a comparatively smaller positive influence and **Superheat** shows only a weak negative relationship.

Existing Dataset (April)	Trial Dataset
AVG_CAST_SPEED → 0.738	CAST_SPEED_ACT → 0.726
MOLD_LEV → -0.327	MLF → 0.104
SUPERHEAT → -0.073	SUPERHEAT → -0.081

Observation

- Casting Speed remains the dominant process parameter in both the existing and trial datasets, exhibiting the highest positive correlation with Average Heat Flux.
- Mould Level behaves differently in the two datasets. In the April dataset, Mould Level shows a moderate negative correlation ($r = -0.327$), whereas in the trial dataset Mould Level Fluctuation (MLF) exhibits a weak positive correlation ($r = 0.104$). This indicates that the modified operating conditions changed the influence of mould level on heat transfer.
- Superheat shows a weak negative correlation in both datasets (-0.073 in the existing dataset and -0.081 in the trial dataset), suggesting that its influence on Average Heat Flux remains comparatively small.
- The comparison indicates that although Casting Speed continues to be the most influential operating parameter, the effect of mould level changes under the trial conditions. This behaviour may be attributed to the modified funnel mould and casting conditions used during the industrial trial, which altered the mould-level dynamics while maintaining similar thermal behaviour with respect to superheat.

	1	2	3	4	5	6
1		AVG_HF	CAST_SPEED_ACT	SUPERHEAT	MLF	
2	AVG_HF	1				
3	CAST_SPEED_ACT	0.726343279	1			
4	SUPERHEAT	-0.080719175	-0.179530204	1		
5	MLF	0.104321178	-0.137874103	-0.021050858	1	
6						

Fig 5.0 Correlation Analysis for the Industrial Trial DataSet

5.4 Multiple Linear Regression Analysis of Trial Dataset

Multiple Linear Regression (MLR) was performed to quantify the combined influence of the selected process parameters on **Average Heat Flux (AVG_HF)** during the industrial trial. The dependent variable considered in this study was **Average Heat Flux (AVG_HF)**, while **Actual Casting Speed (CAST_SPEED_ACT)**, **Superheat (SUPERHEAT)**, and **Mould Level Fluctuation (MLF)** were selected as the independent variables. The regression analysis was performed using **Microsoft Excel**.

The general form of the regression equation is:

$$AVG_HF = \beta_0 + \beta_1(CAST_SPEED_ACT) + \beta_2(SUPERHEAT) + \beta_3(MLF)$$

where:

- **AVG_HF** = Average Heat Flux
- β_0 = Intercept
- β_1 – β_3 = Regression coefficients of the selected process parameters

The regression model produced a **Multiple R value of 0.758**, indicating a strong correlation between the observed and predicted values of Average Heat Flux. The **R² value of 0.574** indicates that approximately **57.4%** of the variation in Average Heat Flux is explained by the selected process parameters. The **Adjusted R² value of 0.574** further confirms the reliability of the regression model after accounting for the number of predictor variables.

The ANOVA results indicate that the regression model is statistically significant, with a **Significance F value close to zero**, demonstrating that the selected independent variables collectively have a significant influence on Average Heat Flux.

Among the regression coefficients, **MLF** has the highest positive coefficient (**0.5087**), indicating that mould level fluctuation has the strongest contribution to the predicted Average Heat Flux under the analysed trial conditions. **Actual Casting Speed** also has a significant positive coefficient (**0.3423**), confirming that higher casting speeds increase heat extraction during continuous casting. **Superheat** has a comparatively smaller positive coefficient (**0.0033**), indicating that its influence on Average Heat Flux is relatively limited within the trial dataset.

The **p-values** of all three predictor variables are significantly smaller than **0.05**, confirming that **CAST_SPEED_ACT**, **SUPERHEAT**, and **MLF** are statistically significant predictors of Average Heat Flux.

Overall, the Multiple Linear Regression analysis demonstrates that the selected process parameters successfully explain the variation in Average Heat Flux during the industrial trial, with **MLF** and **Actual Casting Speed** contributing most strongly to the regression model.

Table 5.2 Regression Model Summary

Parameter	Trial Dataset
Multiple R	0.758
R Square	0.574
Adjusted R Square	0.574
Standard Error	0.153
Observations	63,999

Table 5.3 Regression Coefficients

Parameter	Coefficient Effect	
Intercept	0.2593	Constant
CAST_SPEED_ACT	0.3423	Positive
SUPERHEAT	0.00329	Positive
MLF	0.5087	Positive

1	SUMMARY OUTPUT										
2											
3	Regression Statistics										
4	Multiple R	0.757511361									
5	R Square	0.573823461									
6	Adjusted R Square	0.573803483									
7	Standard Error	0.153158327									
8	Observations	63999									
9											
10	ANOVA										
11		df	SS	MS	F	Significance F					
12	Regression	3	2021.231391	673.7437971	28721.92553	0					
13	Residual	63995	1501.160995	0.023457473							
14	Total	63998	3522.392386								
15											
16		Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%		
17	Intercept	0.259298217	0.008502494	30.49672388	8.2414E-203	0.242633319	0.275963114	0.242633319	0.275963114		
18	CAST_SPEED_ACT	0.342323452	0.001183913	289.1458225	0	0.340002982	0.344643923	0.340002982	0.344643923		
19	SUPERHEAT	0.00329236	0.000140964	23.35609942	3.8123E-120	0.003016071	0.003568649	0.003016071	0.003568649		
20	MLF	0.508727588	0.006280207	81.0049041	0	0.496418376	0.521036801	0.496418376	0.521036801		
21											
22											
23											
24	RESIDUAL OUTPUT										
25											
26	Observation	Predicted AVG_HF	Residuals								
27	1	2.432425483	-0.135175483								

Figure 5.1: Multiple Linear Regression Summary Output for the Industrial Trial Dataset

5.5 Random Forest Analysis

Random Forest Regression was employed to determine the relative importance of the selected process parameters in predicting **Average Heat Flux (AVG_HF)** during the industrial trial. Unlike Multiple Linear Regression, Random Forest is capable of modelling nonlinear relationships and interactions among variables. The model was trained using **Actual Casting Speed (CAST_SPEED_ACT)**, **Superheat (SUPERHEAT)**, and **Mould Level Fluctuation (MLF)** as input variables, while **Average Heat Flux (AVG_HF)** was considered as the response variable.

The feature importance values obtained from the Random Forest model reveal that **MLF** is the most influential parameter with an importance value of **0.684**, contributing approximately **68.4%** of the overall model importance. This indicates that fluctuations in mould level have the strongest influence on Average Heat Flux under the industrial trial conditions.

The second most significant parameter is **Actual Casting Speed**, with a feature importance value of **0.246**, confirming that casting speed continues to play an important role in governing heat transfer during continuous casting.

The influence of **Superheat** is comparatively small, with a feature importance value of **0.070**, indicating that variations in superheat have only a minor contribution towards predicting Average Heat Flux in the trial dataset.

Unlike the historical dataset where casting speed was identified as the dominant process variable, the Random Forest analysis of the trial dataset highlights **Mould Level Fluctuation (MLF)** as the primary controlling parameter. This shift suggests that the modified operating conditions introduced during the industrial trial increased the sensitivity of heat transfer to mould level fluctuations.

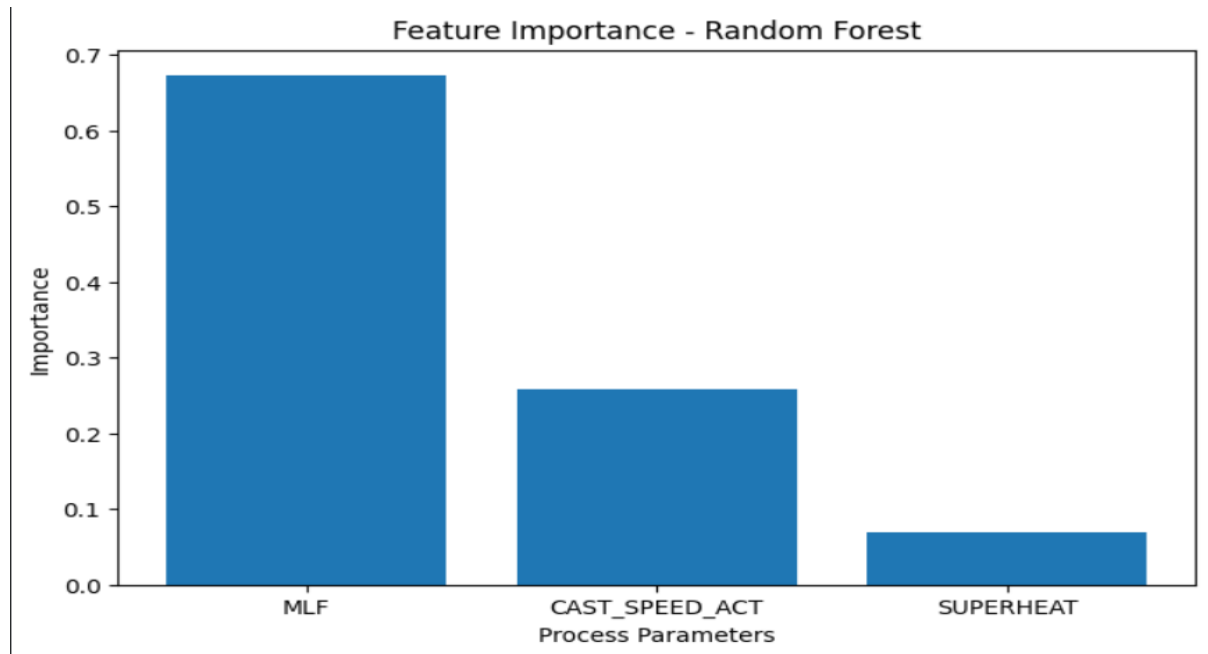


Figure 5.2: Feature Importance from Random Forest Analysis (Industrial Trial Dataset)

Note: Table 5.4 Feature Importance Obtained from Random Forest

Rank	Process Parameter	Importance
1	MLF	0.684
2	CAST_SPEED_ACT	0.246
3	SUPERHEAT	0.070

5.6 Comparison Between Existing Process and Industrial Trial Results

To evaluate the influence of the modified funnel mould and operating conditions, the statistical and machine learning analyses obtained from the industrial trial were compared with those of the historical production dataset. Correlation Analysis, Multiple Linear Regression, and Random Forest Regression were employed to investigate changes in the relationship between process parameters and Average Heat Flux.

The correlation analysis revealed that **Casting Speed** remained the strongest positively correlated parameter in both datasets. However, the effect of mould level changed considerably. In the existing dataset, **Mould Level** exhibited a moderate negative correlation with Average Heat Flux ($r = -0.327$), whereas in the industrial trial **MLF** showed a weak positive correlation

($r = 0.104$). This indicates that the modified operating conditions altered the influence of mould level on heat transfer.

The regression analysis shows that the historical dataset produced a slightly stronger regression model ($R^2 = 0.570$) compared with the industrial trial ($R^2 = 0.574$). Although the improvement is modest, both models demonstrate that the selected process parameters explain a significant portion of the variation in Average Heat Flux.

The Random Forest analysis highlights the most notable difference between the two datasets. In the historical dataset, **Average Casting Speed** was identified as the most influential process parameter, whereas the industrial trial identified **Mould Level Fluctuation (MLF)** as the dominant factor with a feature importance of **68.4%**. This shift suggests that the modified mould conditions increased the influence of mould level behaviour on heat transfer.

Overall, the comparison demonstrates that although **Casting Speed** remains an important operating parameter, the industrial trial shifted the dominant controlling factor from casting speed to **Mould Level Fluctuation**, indicating a change in the heat transfer characteristics under the modified process conditions.

Table 5.5 Comparison Between Existing and Trial Dataset Results

Parameter	Existing Process	Industrial Trial
Multiple R	0.754	0.758
R Square	0.570	0.574
Adjusted R Square	0.569	0.574
Standard Error	0.148	0.153
Most Important Parameter (Random Forest)	AVG_CAST_SPEED	MLF
Second Most Important Parameter	MOLD_LEV	CAST_SPEED_ACT

Key Findings

- Casting Speed remained an important parameter in both the historical and industrial trial datasets.
- The industrial trial identified **Mould Level Fluctuation (MLF)** as the dominant factor influencing Average Heat Flux.
- Superheat exhibited only a minor influence in both Multiple Linear Regression and Random Forest analyses.

CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusions

The present study investigated the influence of important process parameters on Average Heat Flux (AVG_HF) during the Thin Slab Casting and Rolling (TSCR) process using both the existing production dataset and an industrial trial dataset. Statistical techniques and machine learning approaches were employed to identify the process variables governing mould heat transfer.

Correlation analysis was first performed to evaluate the relationship between the selected process parameters and Average Heat Flux. In the existing production dataset, Average Casting Speed exhibited the strongest positive correlation with Average Heat Flux. Similarly, in the industrial trial dataset, Actual Casting Speed remained the parameter with the highest positive correlation ($r = 0.726$), confirming that casting speed continues to play a major role in determining mould heat transfer. Mould Level Fluctuation (MLF) showed a weak positive correlation ($r = 0.104$), whereas Superheat exhibited only a weak negative correlation ($r = -0.081$).

Multiple Linear Regression (MLR) analysis was carried out to quantify the combined influence of the selected process parameters on Average Heat Flux. The industrial trial regression model achieved a Multiple R value of 0.758 and an R^2 value of 0.574, indicating that approximately 57.4% of the variation in Average Heat Flux is explained by the selected process parameters. The regression coefficients revealed that Actual Casting Speed and Mould Level Fluctuation positively influence Average Heat Flux, while the effect of Superheat was comparatively small.

Random Forest Regression was employed to determine the relative importance of the selected process parameters. Unlike the historical dataset, where Average Casting Speed was identified as the most influential parameter, the industrial trial dataset showed Mould Level Fluctuation (MLF) as the dominant parameter with a feature importance of 68.4%, followed by Actual Casting Speed (24.6%) and Superheat (7.0%). This indicates that under the modified trial operating conditions, mould level fluctuations became the primary factor affecting heat transfer behaviour.

A comparison between the existing production data and the industrial trial demonstrates that Casting Speed remained an important parameter under both operating conditions. However, the industrial trial altered the influence of mould level, resulting in MLF becoming the dominant parameter in the Random Forest model. This shift suggests that the modified operating conditions influenced the thermal behaviour of the mould and changed the relative contribution of process variables governing Average Heat Flux.

Overall, the combined application of Correlation Analysis, Multiple Linear Regression, and Random Forest Regression successfully identified the major process parameters affecting Average Heat Flux during continuous casting. The study demonstrates that integrating statistical methods with machine learning techniques provides a reliable approach for analysing process behaviour and identifying critical operating parameters for improving casting performance and process stability.

6.2 Future Scope

The present study demonstrates that statistical analysis and machine learning techniques can effectively identify the important process parameters influencing Average Heat Flux during continuous casting. However, several opportunities exist for extending this work in future.

- A larger industrial dataset covering different steel grades, casting speeds, and production conditions can be analysed to improve the robustness and generalization of the developed models.
- Additional process parameters such as mould cooling water flow rate, mould powder properties, oscillation frequency, and casting temperature may be incorporated to obtain a more comprehensive understanding of mould heat transfer.
- Advanced machine learning algorithms such as **XGBoost, Gradient Boosting, Support Vector Regression (SVR), and Artificial Neural Networks (ANN)** can be implemented and compared with Random Forest to improve prediction accuracy.
- Real-time monitoring and prediction of Average Heat Flux can be developed by integrating machine learning models with plant data acquisition systems, enabling early detection of abnormal casting conditions and reducing breakout risk.
- The developed prediction models can be incorporated into intelligent process control systems for automatic optimization of casting parameters, thereby improving product quality, process stability, and energy efficiency.
- Future investigations may also focus on validating the developed models under different operating conditions and production campaigns to evaluate their industrial applicability and long-term reliability.

Major Outcomes of the Internship

The major outcomes achieved during the industrial internship are summarized below:

- Gained practical exposure to the **Thin Slab Casting and Rolling (TSCR)** process and developed an understanding of continuous casting operations and mould heat transfer mechanisms.
- Collected, processed, and analysed industrial production data obtained from the TSCR plant using Microsoft Excel and Python.
- Calculated **Average Heat Flux (AVG_HF)** from individual mould heat flux measurements to develop a representative response variable for process analysis.
- Performed **Exploratory Data Analysis (EDA)** to understand the characteristics and distribution of important process variables.
- Conducted **Correlation Analysis** to identify the relationships between Average Heat Flux and selected process parameters.
- Developed **Multiple Linear Regression (MLR)** models using Microsoft Excel to quantify the influence of process variables on Average Heat Flux.
- Implemented **Random Forest Regression** using Python to determine the relative importance of selected process parameters affecting mould heat transfer.
- Compared the results obtained from the historical production dataset with those of the industrial trial dataset to evaluate the influence of modified operating conditions.
- Identified **Casting Speed** as the dominant parameter based on correlation analysis, while **Mould Level Fluctuation (MLF)** emerged as the most influential parameter in the Random Forest model for the industrial trial dataset.
- Improved practical knowledge of industrial data analysis, statistical modelling, machine learning, and interpretation of plant process data for engineering applications.
- Developed technical skills in **Microsoft Excel, Python, Pandas, Scikit-learn, data visualization, regression analysis, and feature importance analysis**.
- Enhanced the ability to interpret industrial data and apply data-driven approaches for improving continuous casting process performance.

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Company Source

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